

STAKEHOLDER QUESTION CATALOGUE 11

Timetabling Team

80 practical questions for evidence-disciplined academic scheduling, room suitability, transitions, disruption handling and lower-impact scenarios

STAKEHOLDER	University timetabling professionals who assemble and test draft teaching patterns while authorised academic, room, employment, accessibility and safety owners retain decision authority.
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QUESTION RANGE	TT-001 to TT-080 across eight categories covering calendars, cohorts, rooms, clashes, aggregate feedback, disruption, fairness and scenarios.
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EVIDENCE MODEL	Current authoritative academic and operational sources plus proposed privacy-preserving aggregate telemetry with verified mapping, timing, quality and permissions. Proposed capability is never presented as current deployment.
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CORE PRINCIPLE

Help the timetabling team compare feasible scheduling options from current authoritative constraints and qualified aggregate evidence. Never create or approve the authoritative timetable, infer individual attendance or performance, rank people, override academic, employment, accessibility or room-owner constraints, or treat telemetry as capacity, access or safety approval.

1. Catalogue scope and design principles

Stakeholder-facing questions support evidence-aware draft decisions; they do not transfer authority from current source owners or competent professionals.

This catalogue describes what a Talking Abacws service could answer for a timetabling team. All sensing, analytics and integrations are proposed design requirements unless independently verified. The May 2021 floor plan is historical spatial reference only and cannot establish current room use, capacity, access, equipment, hazards, services or closures.

DECISION SUPPORT, NOT TIMETABLE AUTHORITY

The service may calculate, compare, diagnose and recommend options. It never creates, approves, publishes or changes the authoritative timetable.

PROPOSED CAPABILITY, NOT CURRENT FACT

Treat sensors, mappings, models and integrations as proposed until deployment, commissioning, permission and current health are verified.

CURRENT CONSTRAINTS BEFORE HISTORY

Use current academic, room, employment, accessibility, service and closure owners. Historical timetables and the May 2021 plan provide context only.

HARD CONSTRAINTS BEFORE PREFERENCES

Preserve academic, contractual, accessibility, capacity, safety and room-owner limits. Analyse preferences separately and never relax them silently.

PRIVACY-PRESERVING AGGREGATE EVIDENCE

Use anonymous aggregate room or route counts only when proportionate. Never reconstruct individual attendance, movement, performance or productivity.

HONEST OPERATION AND SAFE FAILURE

Label lookup, observation, calculation, estimate, forecast, comparison, diagnosis and recommendation correctly. Clarify, qualify, defer or return not assessable.

2. How to read each question record

Every record states the timetable decision, minimum evidence, operation, uncertainty, authority boundary and practical value.

TIMETABLING TEAM QUESTION	A natural question a timetabling professional would ask to make a draft scheduling, room, transition, exception or scenario decision.
SENSORS AND TELEMETRY	Only physically plausible, privacy-preserving proposed evidence. The field states asset or area, coverage, interval or aggregation, commissioning and quality limits.
AUTHORITATIVE SUPPORTING DATA	Current owner-controlled academic, cohort, staff, room, access, equipment, service, closure and emergency sources, with provenance, times, permission and conflicts.
ANALYSIS REQUIRED	The named operation, decision window, minimum evidence, method, controls or baseline, uncertainty and safe failure. Estimates and forecasts are not observations.
ANSWER BOUNDARY	What the service must not infer, disclose, promise, certify, authorise or control, and when it must clarify, qualify, defer or return not assessable.
WHY THIS IS USEFUL	The concrete timetable action supported: narrow options, identify a clash, review a room, escalate a conflict, plan a contingency or compare a scenario.
IMPLEMENTATION READINESS	
R1 - PRACTICAL	Primarily authoritative lookup, verified static information or current owner-issued status. Useful before a full sensing programme exists.
R2 - INTEGRATION-DEPENDENT	Requires current cross-system integration and/or proposed privacy-preserving sensing with verified coverage, timing and quality.
R3 - RESEARCH OR RESTRICTED	Requires validated modelling, fairness or optimisation methods, sensitive permissions, specialist governance or higher-consequence interpretation.
GLOBAL RULE	Distinguish proposed capability from verified fact; preserve authoritative-source precedence, hard constraints, privacy and current permissions; return a qualified or not-assessable answer whenever critical evidence, authority or safety status is unverified.

3. Worked question-to-evidence example

A room-allocation request is translated into explicit evidence obligations before any recommendation is made.

EXAMPLE QUESTION

Which current rooms can host a 160-student lecture next term while meeting authorised capacity, whole-route accessibility and required AV constraints?

1. RESOLVE SESSION REQUIREMENTS

Confirm cohort snapshot, approved layout, duration, AV, route and seating requirements, setup time and capacity margin. Separate hard constraints from preferences.

2. VERIFY CURRENT AUTHORITATIVE DATA

Use current room capacity by layout, booking eligibility, equipment, whole-route access, closures and owner status. Preserve source owner, version and effective time.

3. VERIFY ELIGIBLE TELEMETRY CONTEXT

Room-level aggregate use or environment evidence is optional context only. Require exact mapping, commissioning, aligned clocks, uptime, gaps, drift and current health.

4. PERFORM THE RIGHT OPERATIONS

Apply authorised lookup and hard-constraint filtering first; calculate capacity margin and transitions; compare only eligible rooms; report uncertainty and excluded evidence.

5. PRESERVE AUTHORITY AND ALTERNATIVES

Do not grant a room, reinterpret capacity or certify accessibility. Keep verified alternatives and identify the responsible room, equipment and accessibility owners.

6. COMMUNICATE ANSWERABILITY

Separate confirmed facts, calculations and qualified context. If capacity, route, equipment or permission evidence is unverified, return not assessable.

ANSWER SHAPE

Lead with the eligible room set or state that none is assessable. Then show authorised capacity, route and equipment status, calculations, qualified aggregate context, trade-offs, uncertainty, recheck time and source owners.

1. Academic calendar, teaching patterns and scheduling constraints

TT-001 to TT-002 | 10 questions in this section | Resolve valid dates, recurrence, sequence and hard scheduling limits before constructing options.

TT-001

VALID TEACHING WEEKS

L1 - lookup

R1 - practical

TIMETABLING TEAM QUESTION Which dates in the approved academic calendar can this module's recurring teaching use, and which dates are exceptions?

SENSORS AND TELEMETRY

No physical sensing is needed for the core result. Where building-open, closure or service status matters, use only the current owner-issued feed for the named asset, with verified mapping, observed/retrieved time, heartbeat, maintenance state and conflicts. Telemetry does not create permission or availability.

AUTHORITATIVE SUPPORTING DATA

Current academic-calendar owner; approved teaching, assessment and reading weeks; public holidays; institutional closure dates; and the authorised module delivery window. Source owners: institutional academic-calendar owner; programme/module owner; central timetabling owner; relevant room or closure owner. Record source version, effective/observed and retrieved times, permission scope, staleness and conflicts. Current owner-issued records prevail over history, inference or telemetry; equal-authority conflicts remain explicit.

ANALYSIS REQUIRED

Operation: authorised lookup plus recurrence calculation. Decision window: annual timetable construction. Expand the approved pattern, remove blocked dates and list exceptions. Minimum evidence: an approved calendar and module window; otherwise return not assessable. Validate version, effective time, permission and conflicts; never infer hard constraints from history or telemetry. If minimum evidence is missing, stale, conflicting or unauthorised, qualify the answer or return not assessable.

ANSWER BOUNDARY

Do not copy dates from a previous year, approve the calendar or publish the timetable. Building telemetry and the May 2021 floor plan cannot define academic dates; unresolved ownership must be deferred. The service cannot create, approve, publish or change the timetable.

WHY THIS IS USEFUL

Establishes the correct session-date set before rooms, cohorts and staff are matched.

TT-002

RECURRING SLOT WINDOWS

L2 - calculation

R1 - practical

TIMETABLING TEAM QUESTION Which recurring two-hour windows remain feasible after applying approved teaching hours, protected breaks, setup time and programme rules?

SENSORS AND TELEMETRY

No physical sensing is needed for the core result. Where building-open, closure or service status matters, use only the current owner-issued feed for the named asset, with verified mapping, observed/retrieved time, heartbeat, maintenance state and conflicts. Telemetry does not create permission or availability.

AUTHORITATIVE SUPPORTING DATA

Current teaching-day policy; module duration and recurrence; programme constraints; staff-declared scheduling windows; protected breaks; setup or clearance time; and approved local rules. Source owners: institutional academic-calendar owner; programme/module owner; central timetabling owner; relevant room or closure owner. Record source version, effective/observed and retrieved times, permission scope, staleness and conflicts. Current owner-issued records prevail over history, inference or telemetry; equal-authority conflicts remain explicit.

ANALYSIS REQUIRED

Operation: constraint calculation. Decision window: draft timetable cycle. Generate only windows satisfying every hard rule, then display preferences separately. Missing or conflicting rules require clarification rather than assumption. Validate version, effective time, permission and conflicts; never infer hard constraints from history or telemetry. If minimum evidence is missing, stale, conflicting or unauthorised, qualify the answer or return not assessable.

ANSWER BOUNDARY

Do not decide contractual availability, reasonable adjustments or academic priorities, and do not infer availability from a personal calendar beyond authorised scheduling fields. Timetablers choose among feasible windows. The service cannot create, approve, publish or change the timetable.

WHY THIS IS USEFUL

Narrows the search to usable recurring patterns and prevents avoidable rule breaches.

1. Academic calendar, teaching patterns and scheduling constraints

TT-003 to TT-004 | 10 questions in this section | Resolve valid dates, recurrence, sequence and hard scheduling limits before constructing options.

TT-003

WEEKLY OR BLOCK DELIVERY

L3 - comparison

R2 - integration-dependent

TIMETABLING TEAM QUESTION

Is weekly delivery or an approved intensive block more feasible for this module, and what trade-offs does each option create?

SENSORS AND TELEMETRY

No physical sensing is needed for the core result. Where building-open, closure or service status matters, use only the current owner-issued feed for the named asset, with verified mapping, observed/retrieved time, heartbeat, maintenance state and conflicts. Telemetry does not create permission or availability.

AUTHORITATIVE SUPPORTING DATA

Approved module delivery modes and contact hours; cohort commitments; staff availability; room eligibility; assessment pattern; setup needs; programme policy; and publication deadlines. Source owners: institutional academic-calendar owner; programme/module owner; central timetabling owner; relevant room or closure owner. Record source version, effective/observed and retrieved times, permission scope, staleness and conflicts. Current owner-issued records prevail over history, inference or telemetry; equal-authority conflicts remain explicit.

ANALYSIS REQUIRED

Operation: constrained scenario comparison. Decision window: module-pattern design. Build both authorised patterns, compare clashes, room demand, day length and transition burden, and report sensitivity to uncertain enrolment or room availability. If block delivery is not approved, exclude it. Validate version, effective time, permission and conflicts; never infer hard constraints from history or telemetry. Compare only eligible cases under matched programme, room, time and occupancy conditions; disclose residual confounders.

ANSWER BOUNDARY

Do not change the module delivery mode, compress required learning activity or treat a modelled pattern as academic approval. The module and programme owners decide whether an alternative pattern is permissible. The service cannot create, approve, publish or change the timetable.

WHY THIS IS USEFUL

Shows whether a different authorised pattern genuinely increases feasibility before detailed allocation.

TT-004

BLOCKED AND RESERVED DATES

L1 - lookup

R1 - practical

TIMETABLING TEAM QUESTION

Which proposed teaching dates are blocked by assessment, reading-week, closure or institution-wide event rules?

SENSORS AND TELEMETRY

No physical sensing is needed for the core result. Where building-open, closure or service status matters, use only the current owner-issued feed for the named asset, with verified mapping, observed/retrieved time, heartbeat, maintenance state and conflicts. Telemetry does not create permission or availability.

AUTHORITATIVE SUPPORTING DATA

Current academic calendar; examination and assessment windows; reading weeks; owner-issued closures; institution-wide events; and programme-specific reserved dates. Source owners: institutional academic-calendar owner; programme/module owner; central timetabling owner; relevant room or closure owner. Record source version, effective/observed and retrieved times, permission scope, staleness and conflicts. Current owner-issued records prevail over history, inference or telemetry; equal-authority conflicts remain explicit.

ANALYSIS REQUIRED

Operation: authorised lookup plus set intersection. Decision window: the proposed occurrence set. Match each date against active restrictions and label the owning rule. If two owners disagree, expose the conflict and return the affected dates as unresolved. Validate version, effective time, permission and conflicts; never infer hard constraints from history or telemetry. If minimum evidence is missing, stale, conflicting or unauthorised, qualify the answer or return not assessable.

ANSWER BOUNDARY

Do not infer restrictions from historical patterns or building occupancy, and do not decide which authoritative rule should be relaxed. Only the responsible owner can amend a block. The service cannot create, approve, publish or change the timetable.

WHY THIS IS USEFUL

Prevents sessions being placed on dates that are unavailable or reserved for another purpose.

1. Academic calendar, teaching patterns and scheduling constraints

TT-005 to TT-006 | 10 questions in this section | Resolve valid dates, recurrence, sequence and hard scheduling limits before constructing options.

TT-005

HOLIDAY REPLACEMENT SESSIONS

L2 - calculation

R1 - practical

TIMETABLING TEAM QUESTION

After holidays and one-off closures are applied, which sessions need an authorised replacement date to preserve the required contact pattern?

SENSORS AND TELEMETRY

No physical sensing is needed for the core result. Where building-open, closure or service status matters, use only the current owner-issued feed for the named asset, with verified mapping, observed/retrieved time, heartbeat, maintenance state and conflicts. Telemetry does not create permission or availability.

AUTHORITATIVE SUPPORTING DATA

Approved session recurrence and contact-hour requirement; holiday and closure dates; module delivery window; replacement-session policy; and programme restrictions. Source owners: institutional academic-calendar owner; programme/module owner; central timetabling owner; relevant room or closure owner. Record source version, effective/observed and retrieved times, permission scope, staleness and conflicts. Current owner-issued records prevail over history, inference or telemetry; equal-authority conflicts remain explicit.

ANALYSIS REQUIRED

Operation: recurrence calculation and gap identification. Decision window: the module delivery period. Count cancelled occurrences, test only authorised replacement windows and show any remaining shortfall. If the contact requirement is ambiguous, ask the module owner. Validate version, effective time, permission and conflicts; never infer hard constraints from history or telemetry. If minimum evidence is missing, stale, conflicting or unauthorised, qualify the answer or return not assessable.

ANSWER BOUNDARY

Do not invent replacement teaching, alter contact hours or assume a cancelled session must be rescheduled. The service identifies the gap; academic owners decide the remedy. The service cannot create, approve, publish or change the timetable.

WHY THIS IS USEFUL

Makes holiday effects visible early enough to secure a valid replacement pattern.

TT-006

ACTIVITY RECURRENCE TYPE

L2 - calculation

R1 - practical

TIMETABLING TEAM QUESTION

Which activities must run weekly, fortnightly, in approved blocks or once under the current module delivery plan?

SENSORS AND TELEMETRY

No physical sensing is needed for the core result. Where building-open, closure or service status matters, use only the current owner-issued feed for the named asset, with verified mapping, observed/retrieved time, heartbeat, maintenance state and conflicts. Telemetry does not create permission or availability.

AUTHORITATIVE SUPPORTING DATA

Current module specification; activity catalogue; approved contact pattern; group structure; delivery dates; dependencies; and programme-owner instructions. Source owners: institutional academic-calendar owner; programme/module owner; central timetabling owner; relevant room or closure owner. Record source version, effective/observed and retrieved times, permission scope, staleness and conflicts. Current owner-issued records prevail over history, inference or telemetry; equal-authority conflicts remain explicit.

ANALYSIS REQUIRED

Operation: authorised lookup plus pattern classification. Decision window: module setup. Map each activity to its approved recurrence and generate occurrence counts. Unmapped or contradictory activities are returned for owner clarification. Validate version, effective time, permission and conflicts; never infer hard constraints from history or telemetry. If minimum evidence is missing, stale, conflicting or unauthorised, qualify the answer or return not assessable.

ANSWER BOUNDARY

Do not infer recurrence from an activity name, prior timetable or lecturer preference, and do not create new activity types. Only approved module data defines the pattern. The service cannot create, approve, publish or change the timetable.

WHY THIS IS USEFUL

Ensures the timetable models each teaching activity with the correct recurrence from the outset.

1. Academic calendar, teaching patterns and scheduling constraints

TT-007 to TT-008 | 10 questions in this section | Resolve valid dates, recurrence, sequence and hard scheduling limits before constructing options.

TT-007

REQUIRED ACTIVITY SEQUENCE

L3 - recommendation

R2 - integration-dependent

TIMETABLING TEAM QUESTION

Can the lecture, seminar and practical be placed in the approved order with the required intervals between them?

SENSORS AND TELEMETRY

No physical sensing is needed for the core result. Where building-open, closure or service status matters, use only the current owner-issued feed for the named asset, with verified mapping, observed/retrieved time, heartbeat, maintenance state and conflicts. Telemetry does not create permission or availability.

AUTHORITATIVE SUPPORTING DATA

Approved activity dependencies; minimum and maximum intervals; cohort and staff constraints; eligible room windows; teaching weeks; and programme-owner exceptions. Source owners: institutional academic-calendar owner; programme/module owner; central timetabling owner; relevant room or closure owner. Record source version, effective/observed and retrieved times, permission scope, staleness and conflicts. Current owner-issued records prevail over history, inference or telemetry; equal-authority conflicts remain explicit.

ANALYSIS REQUIRED

Operation: precedence-constrained calculation followed by recommendation. Decision window: each activity sequence. Find feasible ordered combinations, quantify gaps and report the smallest relaxation needed if none exist. Do not relax a rule automatically. Validate version, effective time, permission and conflicts; never infer hard constraints from history or telemetry. Filter hard constraints first, compare only eligible options under explicit criteria, show alternatives and never apply changes automatically.

ANSWER BOUNDARY

Do not infer pedagogy or sequence from titles, and do not override the module owner's dependency rules. If the required interval is missing or disputed, return not assessable. The service cannot create, approve, publish or change the timetable.

WHY THIS IS USEFUL

Lets the team protect academic sequencing before individual rooms are allocated.

TT-008

LONG OR FRAGMENTED DAYS

L2 - comparison

R2 - integration-dependent

TIMETABLING TEAM QUESTION

Which cohorts have the longest or most fragmented days once realistic travel and usable breaks are included?

SENSORS AND TELEMETRY

No physical sensing is needed for the core schedule comparison. Proposed anonymous directional counters may qualify recurrent congestion only at the exact corridor, stair or lift-lobby boundaries used by the route, using 1-minute observations and 5-minute aggregates. Require commissioned coverage, synchronized clocks, uptime/gap, drift and relocation status; no identities or path histories.

AUTHORITATIVE SUPPORTING DATA

Current draft timetable; cohort membership at aggregate level; session start and finish times; verified room locations; transition allowances; protected-break rules; and current route or lift status where relevant. Source owners: institutional academic-calendar owner; programme/module owner; central timetabling owner; relevant room or closure owner. Record source version, effective/observed and retrieved times, permission scope, staleness and conflicts. Current owner-issued records prevail over history, inference or telemetry; equal-authority conflicts remain explicit.

ANALYSIS REQUIRED

Operation: timetable-span and fragmentation comparison. Decision window: each teaching day. Calculate first-to-last span, taught hours, transition time and usable gaps; compare only like cohorts and report unresolved route assumptions. Exclude telemetry that fails mapping, clock, coverage or health checks; state excluded intervals and residual uncertainty. Compare only eligible cases under matched programme, room, time and occupancy conditions; disclose residual confounders.

ANSWER BOUNDARY

Do not infer individual attendance, hardship or preference, and do not rank students or programmes. Aggregate results support review; authorised programme owners decide acceptable patterns. Do not substitute an uncovered route segment or adjacent boundary for complete route evidence. The service cannot create, approve, publish or change the timetable.

WHY THIS IS USEFUL

Identifies draft patterns that may need consolidation, redistribution or a larger transition allowance.

1. Academic calendar, teaching patterns and scheduling constraints

TT-009 to TT-010 | 10 questions in this section | Resolve valid dates, recurrence, sequence and hard scheduling limits before constructing options.

TT-009

NO-FEASIBLE-SLOT DIAGNOSIS

L3 - diagnosis

R2 - integration-dependent

TIMETABLING TEAM QUESTION

Which sessions still have no feasible slot, and which hard constraint or unresolved source conflict blocks each one?

SENSORS AND TELEMETRY

No physical sensing is needed for the core result. Where building-open, closure or service status matters, use only the current owner-issued feed for the named asset, with verified mapping, observed/retrieved time, heartbeat, maintenance state and conflicts. Telemetry does not create permission or availability.

AUTHORITATIVE SUPPORTING DATA

Current module, cohort, staff, room, access, equipment, calendar and policy constraints; constraint ownership; precedence; and the current draft. Source owners: institutional academic-calendar owner; programme/module owner; central timetabling owner; relevant room or closure owner. Record source version, effective/observed and retrieved times, permission scope, staleness and conflicts. Current owner-issued records prevail over history, inference or telemetry; equal-authority conflicts remain explicit.

ANALYSIS REQUIRED

Operation: constraint-satisfaction diagnosis. Decision window: unresolved session placement. Identify a minimal conflicting constraint set, distinguish missing data from genuine infeasibility and show authorised owners who could clarify each rule. If the solver or inputs are incomplete, return not assessable. Validate version, effective time, permission and conflicts; never infer hard constraints from history or telemetry. Identify the minimum conflict set or temporal association; do not claim causation.

ANSWER BOUNDARY

Do not choose which hard constraint to break, disclose restricted personal reasons or create an authoritative timetable. The diagnosis explains the conflict; responsible owners resolve it. It may report an association, not an unverified cause. The service cannot create, approve, publish or change the timetable.

WHY THIS IS USEFUL

Turns an opaque scheduling failure into a traceable set of constraints that can be escalated.

TT-010

PREFERENCE SENSITIVITY

L3 - diagnosis

R3 - research or restricted

TIMETABLING TEAM QUESTION

Which optional preferences remove the most otherwise-feasible timetable options?

SENSORS AND TELEMETRY

No physical sensing is needed for the core result. Where building-open, closure or service status matters, use only the current owner-issued feed for the named asset, with verified mapping, observed/retrieved time, heartbeat, maintenance state and conflicts. Telemetry does not create permission or availability.

AUTHORITATIVE SUPPORTING DATA

Approved hard constraints and separately labelled preferences; preference owner, priority and scope; current feasible-option set; and governance rules for sensitive availability. Source owners: institutional academic-calendar owner; programme/module owner; central timetabling owner; relevant room or closure owner. Record source version, effective/observed and retrieved times, permission scope, staleness and conflicts. Current owner-issued records prevail over history, inference or telemetry; equal-authority conflicts remain explicit.

ANALYSIS REQUIRED

Operation: sensitivity diagnosis. Decision window: draft optimisation. Recompute feasible counts with each preference removed in turn, report marginal impact and interactions, and suppress results that could reveal a person's circumstances. If hard and soft rules are not clearly labelled, stop. Validate version, effective time, permission and conflicts; never infer hard constraints from history or telemetry. Identify the minimum conflict set or temporal association; do not claim causation.

ANSWER BOUNDARY

Do not rank staff or students, infer reasons for a preference, pressure an individual to relax it or convert a preference into a hard rule. Authorised owners decide any change. It may report an association, not an unverified cause. The service cannot create, approve, publish or change the timetable.

WHY THIS IS USEFUL

Shows where clarification could recover options without weakening genuine academic, employment or accessibility constraints.

2. Modules, cohorts, staff availability and required activities

TT-011 to TT-012 | 10 questions in this section | Combine current curriculum, aggregate cohort, authorised staff and activity dependencies without exposing personal reasons.

TT-011

SHARED-COHORT CLASHES

L2 - calculation

R1 - practical

TIMETABLING TEAM QUESTION Which modules share an authorised cohort and therefore require clash protection in the same teaching period?

SENSORS AND TELEMETRY

No physical sensing is needed for the core result. Where building-open, closure or service status matters, use only the current owner-issued feed for the named asset, with verified mapping, observed/retrieved time, heartbeat, maintenance state and conflicts. Telemetry does not create permission or availability.

AUTHORITATIVE SUPPORTING DATA

Current programme structures; compulsory-module mapping; authorised cohort membership or aggregate pathway rules; module instances; teaching periods; and approved clash-protection policy. Source owners: programme/module owner; student-records owner for aggregate cohort data; authorised employment/workload owner; specialist room or support owner. Record source version, effective/observed and retrieved times, permission scope, staleness and conflicts. Current owner-issued records prevail over history, inference or telemetry; equal-authority conflicts remain explicit.

ANALYSIS REQUIRED

Operation: authorised lookup plus graph-overlap calculation. Decision window: timetable build. Create module pairs sharing a protected cohort, label the rule and distinguish exact from potential overlap. Missing pathway mapping is reported as a coverage gap. Validate version, effective time, permission and conflicts; never infer hard constraints from history or telemetry. If minimum evidence is missing, stale, conflicting or unauthorised, qualify the answer or return not assessable.

ANSWER BOUNDARY

Do not expose individual enrolments or infer choices from attendance data. The service identifies protected module relationships; programme owners approve the clash matrix. The service cannot create, approve, publish or change the timetable.

WHY THIS IS USEFUL

Provides a defensible clash-protection set before sessions are placed.

TT-012

PARALLEL GROUP COUNT

L3 - estimate

R2 - integration-dependent

TIMETABLING TEAM QUESTION How many parallel seminar or tutorial groups should we plan from the authorised enrolment snapshot and group-size rule?

SENSORS AND TELEMETRY

No physical sensing is needed for the group-count calculation. Proposed anonymous whole-room counts may be used only as secondary historical context, with complete entrance coverage, commissioned error bounds, synchronized clocks, 1-minute observations aggregated over 15 minutes, and uptime/gap, drift and relocation status. Counts never replace authorised enrolment, capacity or booking records.

AUTHORITATIVE SUPPORTING DATA

Current enrolment snapshot and census date; approved maximum and target group sizes; expected late-change allowance; activity recurrence; available teaching staff and eligible rooms. Source owners: programme/module owner; student-records owner for aggregate cohort data; authorised employment/workload owner; specialist room or support owner. Record source version, effective/observed and retrieved times, permission scope, staleness and conflicts. Current owner-issued records prevail over history, inference or telemetry; equal-authority conflicts remain explicit.

ANALYSIS REQUIRED

Operation: calculation plus scenario estimate. Decision window: group-structure planning. Compute the minimum group count, test an authorised enrolment range and show capacity margin. If the enrolment snapshot or group-size rule is provisional, label the result accordingly. Exclude telemetry that fails mapping, clock, coverage or health checks; state excluded intervals and residual uncertainty. Use eligible comparators, report ranges and sensitivity, and return not assessable when coverage or drift is inadequate.

ANSWER BOUNDARY

Do not infer individual attendance, decide pedagogical group size or promise staff or rooms. The module owner approves group structure and resource owners confirm availability. The service cannot create, approve, publish or change the timetable.

WHY THIS IS USEFUL

Supports early creation of a viable number of groups without hiding enrolment uncertainty.

2. Modules, cohorts, staff availability and required activities

TT-013 to TT-014 | 10 questions in this section | Combine current curriculum, aggregate cohort, authorised staff and activity dependencies without exposing personal reasons.

TT-013

AUTHORISED STAFF WINDOWS

L1 - lookup

R3 - research or restricted

TIMETABLING TEAM QUESTION Which teaching windows are currently authorised for each staff member assigned to this activity?

SENSORS AND TELEMETRY

No physical sensing is needed for the core result. Where building-open, closure or service status matters, use only the current owner-issued feed for the named asset, with verified mapping, observed/retrieved time, heartbeat, maintenance state and conflicts. Telemetry does not create permission or availability.

AUTHORITATIVE SUPPORTING DATA

Restricted scheduling-availability fields from the authorised staff system; assigned teaching roles; employment and approved adjustment constraints represented only as usable windows; and data-access permission. Source owners: programme/module owner; student-records owner for aggregate cohort data; authorised employment/workload owner; specialist room or support owner. Record source version, effective/observed and retrieved times, permission scope, staleness and conflicts. Current owner-issued records prevail over history, inference or telemetry; equal-authority conflicts remain explicit.

ANALYSIS REQUIRED

Operation: permission-bounded authorised lookup. Decision window: session placement. Return usable windows and conflict flags only, not reasons. If the requester lacks permission or the assignment is not confirmed, defer to the responsible staffing owner. Validate version, effective time, permission and conflicts; never infer hard constraints from history or telemetry. If minimum evidence is missing, stale, conflicting or unauthorised, qualify the answer or return not assessable.

ANSWER BOUNDARY

Do not inspect personal calendars beyond authorised scheduling constraints, disclose adjustment or employment details, infer willingness, or rank staff. Availability data does not approve a session. The service cannot create, approve, publish or change the timetable.

WHY THIS IS USEFUL

Lets the team test feasible placements while protecting sensitive staff information.

TT-014

CO-TEACHING INTERSECTION

L2 - calculation

R2 - integration-dependent

TIMETABLING TEAM QUESTION Is there a common authorised window for all required co-teachers once cohort and room constraints are applied?

SENSORS AND TELEMETRY

No physical sensing is needed for the core result. Where building-open, closure or service status matters, use only the current owner-issued feed for the named asset, with verified mapping, observed/retrieved time, heartbeat, maintenance state and conflicts. Telemetry does not create permission or availability.

AUTHORITATIVE SUPPORTING DATA

Confirmed co-teaching assignment; permission-bounded staff availability; cohort commitments; eligible room windows; session duration; and required setup or transition time. Source owners: programme/module owner; student-records owner for aggregate cohort data; authorised employment/workload owner; specialist room or support owner. Record source version, effective/observed and retrieved times, permission scope, staleness and conflicts. Current owner-issued records prevail over history, inference or telemetry; equal-authority conflicts remain explicit.

ANALYSIS REQUIRED

Operation: set-intersection calculation. Decision window: the co-taught session. Intersect authorised windows, apply cohort and room constraints and list feasible options with constraint provenance. If one assignment or availability source is unresolved, return the partial result as not yet assessable. Validate version, effective time, permission and conflicts; never infer hard constraints from history or telemetry. If minimum evidence is missing, stale, conflicting or unauthorised, qualify the answer or return not assessable.

ANSWER BOUNDARY

Do not expose individual reasons, infer willingness or select a staff member to remove. The service proposes feasible common windows; authorised owners confirm the assignment and placement. The service cannot create, approve, publish or change the timetable.

WHY THIS IS USEFUL

Avoids repeated manual checks across several staff and resource calendars.

2. Modules, cohorts, staff availability and required activities

TT-015 to TT-016 | 10 questions in this section | Combine current curriculum, aggregate cohort, authorised staff and activity dependencies without exposing personal reasons.

TT-015

SPECIALIST SUPPORT WINDOW

L3 - recommendation

R2 - integration-dependent

TIMETABLING TEAM QUESTION

Which windows can support this practical when the specialist room, authorised support staff and required setup are all available?

SENSORS AND TELEMETRY

Use only owner-issued non-content status for each named room or served asset: door/lift state, workstation or AV inventory/health, network/power availability, setup state or maintenance flag as relevant. Preserve asset-to-room mapping, event and retrieval times, heartbeat, maintenance mode, permission and conflicts. Do not inspect screens, files, traffic or teaching content.

AUTHORITATIVE SUPPORTING DATA

Approved practical requirements; current specialist-room eligibility; equipment or service owner status; authorised support rota; staff and cohort windows; setup, reset and safety-clearance times; and current closures. Source owners: programme/module owner; student-records owner for aggregate cohort data; authorised employment/workload owner; specialist room or support owner. Record source version, effective/observed and retrieved times, permission scope, staleness and conflicts. Current owner-issued records prevail over history, inference or telemetry; equal-authority conflicts remain explicit.

ANALYSIS REQUIRED

Operation: multi-resource feasibility calculation followed by recommendation. Decision window: the practical series. Intersect all required resources, include setup and reset, and rank only fully eligible windows by stated preferences. Missing service or safety status makes the option not assessable. Exclude telemetry that fails mapping, clock, coverage or health checks; state excluded intervals and residual uncertainty. Filter hard constraints first, compare only eligible options under explicit criteria, show alternatives and never apply changes automatically.

ANSWER BOUNDARY

Do not assign technical staff, certify equipment or safety, infer room suitability from vacancy, or operate any asset. Resource and safety owners retain approval. The service cannot create, approve, publish or change the timetable.

WHY THIS IS USEFUL

Finds practical slots that are deliverable, not merely empty in the timetable.

TT-016

APPROVED ACTIVITY DEPENDENCIES

L2 - calculation

R1 - practical

TIMETABLING TEAM QUESTION

Which activities have an approved before-or-after dependency, and what interval must the timetable preserve?

SENSORS AND TELEMETRY

No physical sensing is needed for the core result. Where building-open, closure or service status matters, use only the current owner-issued feed for the named asset, with verified mapping, observed/retrieved time, heartbeat, maintenance state and conflicts. Telemetry does not create permission or availability.

AUTHORITATIVE SUPPORTING DATA

Current module specification; approved activity dependencies; minimum and maximum intervals; exception rules; activity instances; and owning academic service. Source owners: programme/module owner; student-records owner for aggregate cohort data; authorised employment/workload owner; specialist room or support owner. Record source version, effective/observed and retrieved times, permission scope, staleness and conflicts. Current owner-issued records prevail over history, inference or telemetry; equal-authority conflicts remain explicit.

ANALYSIS REQUIRED

Operation: authorised lookup plus precedence calculation. Decision window: module pattern construction. Build the dependency graph, detect cycles or missing intervals and return the required ordering. Ambiguous rules are referred to the module owner. Validate version, effective time, permission and conflicts; never infer hard constraints from history or telemetry. If minimum evidence is missing, stale, conflicting or unauthorised, qualify the answer or return not assessable.

ANSWER BOUNDARY

Do not infer sequence from activity names, learning materials or previous timetables, and do not change the dependency. The service represents approved rules only. The service cannot create, approve, publish or change the timetable.

WHY THIS IS USEFUL

Prevents sessions being placed in an academically invalid order.

2. Modules, cohorts, staff availability and required activities

TT-017 to TT-018 | 10 questions in this section | Combine current curriculum, aggregate cohort, authorised staff and activity dependencies without exposing personal reasons.

TT-017

COMPULSORY AND OPTIONAL PROTECTION

L2 - calculation

R1 - practical

TIMETABLING TEAM QUESTION

Which compulsory activities need full clash protection, and which authorised optional combinations need targeted protection?

SENSORS AND TELEMETRY

No physical sensing is needed for the core result. Where building-open, closure or service status matters, use only the current owner-issued feed for the named asset, with verified mapping, observed/retrieved time, heartbeat, maintenance state and conflicts. Telemetry does not create permission or availability.

AUTHORITATIVE SUPPORTING DATA

Current curriculum and pathway rules; compulsory and optional module status; authorised aggregate option combinations; cohort structures; and programme-owner clash policy. Source owners: programme/module owner; student-records owner for aggregate cohort data; authorised employment/workload owner; specialist room or support owner. Record source version, effective/observed and retrieved times, permission scope, staleness and conflicts. Current owner-issued records prevail over history, inference or telemetry; equal-authority conflicts remain explicit.

ANALYSIS REQUIRED

Operation: authorised classification plus conflict calculation. Decision window: clash-matrix creation. Apply full protection to compulsory combinations and policy-defined protection to eligible options; report uncovered or low-confidence combinations. Validate version, effective time, permission and conflicts; never infer hard constraints from history or telemetry. If minimum evidence is missing, stale, conflicting or unauthorised, qualify the answer or return not assessable.

ANSWER BOUNDARY

Do not disclose or predict individual choices, rank options or decide curriculum policy. Small or sensitive aggregate combinations must be suppressed or handled by programme owners. The service cannot create, approve, publish or change the timetable.

WHY THIS IS USEFUL

Produces a more accurate clash matrix than treating every option as either fully protected or ignored.

TT-018

ENROLMENT CHANGE SCENARIOS

L3 - estimate

R2 - integration-dependent

TIMETABLING TEAM QUESTION

If enrolment moves within the approved planning range, which group counts or room allocations become fragile first?

SENSORS AND TELEMETRY

No physical sensing is needed for the core calculation. Proposed anonymous room-level doorway or ceiling counts may be used only as secondary historical evidence, with complete entrance coverage, commissioned error bounds, aligned clocks, 1-minute observations aggregated to 15 minutes, and uptime, gap, drift and relocation flags. Counts never replace enrolment, capacity or booking records.

AUTHORITATIVE SUPPORTING DATA

Current enrolment snapshot; authorised lower and upper planning scenarios; group-size rules; room capacities by layout; module activity pattern; and change deadlines. Source owners: programme/module owner; student-records owner for aggregate cohort data; authorised employment/workload owner; specialist room or support owner. Record source version, effective/observed and retrieved times, permission scope, staleness and conflicts. Current owner-issued records prevail over history, inference or telemetry; equal-authority conflicts remain explicit.

ANALYSIS REQUIRED

Operation: threshold calculation and scenario estimate. Decision window: pre-teaching review. Recalculate groups and room capacity margins for each scenario and identify the first constraint crossed. Report ranges, not a single precise forecast. Exclude telemetry that fails mapping, clock, coverage or health checks; state excluded intervals and residual uncertainty. Use eligible comparators, report ranges and sensitivity, and return not assessable when coverage or drift is inadequate.

ANSWER BOUNDARY

Do not predict named students, promise a room or alter group-size rules. If the planning range or room capacity is unverified, limit the answer to the known threshold. Do not infer individual attendance, absence, engagement, performance or reasons from aggregate use. The service cannot create, approve, publish or change the timetable.

WHY THIS IS USEFUL

Highlights allocations that need contingency options before late enrolment changes arrive.

2. Modules, cohorts, staff availability and required activities

TT-019 to TT-020 | 10 questions in this section | Combine current curriculum, aggregate cohort, authorised staff and activity dependencies without exposing personal reasons.

TT-019

FORTNIGHTLY ROTATION

L3 - recommendation

R2 - integration-dependent

TIMETABLING TEAM QUESTION

Can these activities rotate fortnightly while preserving clash protection, equal access and the approved session total?

SENSORS AND TELEMETRY

No physical sensing is needed for the core result. Where building-open, closure or service status matters, use only the current owner-issued feed for the named asset, with verified mapping, observed/retrieved time, heartbeat, maintenance state and conflicts. Telemetry does not create permission or availability.

AUTHORITATIVE SUPPORTING DATA

Approved rotation rule; group membership at aggregate level; occurrence counts; shared resources; academic calendar; staff and room windows; and any equity or accessibility constraints. Source owners: programme/module owner; student-records owner for aggregate cohort data; authorised employment/workload owner; specialist room or support owner. Record source version, effective/observed and retrieved times, permission scope, staleness and conflicts. Current owner-issued records prevail over history, inference or telemetry; equal-authority conflicts remain explicit.

ANALYSIS REQUIRED

Operation: pattern calculation and constrained recommendation. Decision window: the teaching period. Generate alternating patterns, verify every group receives the approved occurrences and compare clashes and day patterns. If group mapping is incomplete, return not assessable. Validate version, effective time, permission and conflicts; never infer hard constraints from history or telemetry. Filter hard constraints first, compare only eligible options under explicit criteria, show alternatives and never apply changes automatically.

ANSWER BOUNDARY

Do not infer group membership from attendance, create unequal provision or approve a rotation. Academic owners confirm the pattern and any exceptions. The service cannot create, approve, publish or change the timetable.

WHY THIS IS USEFUL

Supports efficient use of shared rooms or staff without losing required teaching access.

TT-020

CROSS-PROGRAMME ELECTIVES

L3 - diagnosis

R2 - integration-dependent

TIMETABLING TEAM QUESTION

Are authorised cross-programme elective combinations creating clashes that the current cohort model misses?

SENSORS AND TELEMETRY

No physical sensing is needed for the core result. Where building-open, closure or service status matters, use only the current owner-issued feed for the named asset, with verified mapping, observed/retrieved time, heartbeat, maintenance state and conflicts. Telemetry does not create permission or availability.

AUTHORITATIVE SUPPORTING DATA

Aggregate authorised option combinations; programme structures; draft timetable; teaching periods; route allowances where relevant; suppression rules for small groups; and clash-protection policy. Source owners: programme/module owner; student-records owner for aggregate cohort data; authorised employment/workload owner; specialist room or support owner. Record source version, effective/observed and retrieved times, permission scope, staleness and conflicts. Current owner-issued records prevail over history, inference or telemetry; equal-authority conflicts remain explicit.

ANALYSIS REQUIRED

Operation: combination-based conflict diagnosis. Decision window: draft review. Test eligible elective pairs and travel sequences, distinguish exact clashes from route risk and report data coverage. Missing or disclosure-sensitive combinations are excluded and labelled. Validate version, effective time, permission and conflicts; never infer hard constraints from history or telemetry. Identify the minimum conflict set or temporal association; do not claim causation.

ANSWER BOUNDARY

Do not expose or predict individual choices, infer interests or rank programmes. Programme owners decide which aggregate combinations merit protection. It may report an association, not an unverified cause. The service cannot create, approve, publish or change the timetable.

WHY THIS IS USEFUL

Finds option clashes that a compulsory-cohort model can miss.

3. Room capacity, layout, equipment and accessibility suitability

TT-021 to TT-022 | 10 questions in this section | Filter by current room eligibility, whole-route access, equipment, service and evidence coverage before comparing preferences.

TT-021

CURRENT CAPACITY MATCH

L2 - calculation

R1 - practical

TIMETABLING TEAM QUESTION

Which currently eligible rooms can hold this cohort in the approved layout with the required capacity margin?

SENSORS AND TELEMETRY

No physical sensing is needed for the core calculation. Proposed anonymous room-level doorway or ceiling counts may be used only as secondary historical evidence, with complete entrance coverage, commissioned error bounds, aligned clocks, 1-minute observations aggregated to 15 minutes, and uptime, gap, drift and relocation flags. Counts never replace enrolment, capacity or booking records.

AUTHORITATIVE SUPPORTING DATA

Current room-inventory owner; capacity for the approved layout; booking eligibility; room purpose; cohort snapshot; required buffer; access restrictions; and closures. Source owners: central room-data and timetabling owners; room owner; IT/AV or specialist-equipment owner; accessibility service; Estates/Facilities for served assets. Record source version, effective/observed and retrieved times, permission scope, staleness and conflicts. Current owner-issued records prevail over history, inference or telemetry; equal-authority conflicts remain explicit.

ANALYSIS REQUIRED

Operation: authorised lookup plus capacity-margin calculation. Decision window: room allocation. Filter by eligibility first, then compute spare capacity for the approved layout. Unverified or conflicting capacity makes the room not assessable. Exclude telemetry that fails mapping, clock, coverage or health checks; state excluded intervals and residual uncertainty. If minimum evidence is missing, stale, conflicting or unauthorised, qualify the answer or return not assessable.

ANSWER BOUNDARY

Do not infer capacity from furniture, occupancy counts or the May 2021 plan; grant a booking; or certify safety. The room owner defines capacity and eligibility. Do not infer individual attendance, absence, engagement, performance or reasons from aggregate use. The service cannot create, approve, publish or change the timetable.

WHY THIS IS USEFUL

Creates a defensible shortlist of rooms that are large enough and currently eligible.

TT-022

LAYOUT AND PEDAGOGY FIT

L2 - comparison

R2 - integration-dependent

TIMETABLING TEAM QUESTION

Which eligible rooms support the approved seminar or group-work layout without breaching capacity or setup limits?

SENSORS AND TELEMETRY

Use only owner-issued non-content status for each named room or served asset: door/lift state, workstation or AV inventory/health, network/power availability, setup state or maintenance flag as relevant. Preserve asset-to-room mapping, event and retrieval times, heartbeat, maintenance mode, permission and conflicts. Do not inspect screens, files, traffic or teaching content.

AUTHORITATIVE SUPPORTING DATA

Current room layouts and capacities; movable-furniture rules; activity requirements; setup and reset time; accessibility provisions; room-owner restrictions; and current booking status. Source owners: central room-data and timetabling owners; room owner; IT/AV or specialist-equipment owner; accessibility service; Estates/Facilities for served assets. Record source version, effective/observed and retrieved times, permission scope, staleness and conflicts. Current owner-issued records prevail over history, inference or telemetry; equal-authority conflicts remain explicit.

ANALYSIS REQUIRED

Operation: hard-constraint filtering plus suitability comparison. Decision window: room selection. Exclude rooms lacking the approved layout or setup window, then compare eligible alternatives. If the current configuration is unverified, state it as an owner check. Exclude telemetry that fails mapping, clock, coverage or health checks; state excluded intervals and residual uncertainty. Compare only eligible cases under matched programme, room, time and occupancy conditions; disclose residual confounders.

ANSWER BOUNDARY

Do not infer layout from historic drawings or apparent vacancy, move furniture or promise room setup. Room and accessibility owners confirm the usable configuration. The service cannot create, approve, publish or change the timetable.

WHY THIS IS USEFUL

Avoids assigning an apparently large room that cannot support the intended teaching format.

3. Room capacity, layout, equipment and accessibility suitability

TT-023 to TT-024 | 10 questions in this section | Filter by current room eligibility, whole-route access, equipment, service and evidence coverage before comparing preferences.

TT-023

COMPUTING ROOM REQUIREMENTS

L3 - recommendation

R2 - integration-dependent

TIMETABLING TEAM QUESTION

Which eligible rooms have the verified workstations, approved software, power and network service required for this activity?

SENSORS AND TELEMETRY

Use only owner-issued non-content status for each named room or served asset: door/lift state, workstation or AV inventory/health, network/power availability, setup state or maintenance flag as relevant. Preserve asset-to-room mapping, event and retrieval times, heartbeat, maintenance mode, permission and conflicts. Do not inspect screens, files, traffic or teaching content.

AUTHORITATIVE SUPPORTING DATA

Current room and workstation inventory; approved software image and licence scope; workstation availability count; network and power service status; room booking eligibility; access; and support ownership. Source owners: central room-data and timetabling owners; room owner; IT/AV or specialist-equipment owner; accessibility service; Estates/Facilities for served assets. Record source version, effective/observed and retrieved times, permission scope, staleness and conflicts. Current owner-issued records prevail over history, inference or telemetry; equal-authority conflicts remain explicit.

ANALYSIS REQUIRED

Operation: authorised lookup, resource calculation and recommendation. Decision window: room allocation. Match the cohort and contingency margin to verified workstations, then require all named services. If software, network or workstation status is stale, defer to IT or return not assessable. Exclude telemetry that fails mapping, clock, coverage or health checks; state excluded intervals and residual uncertainty. Filter hard constraints first, compare only eligible options under explicit criteria, show alternatives and never apply changes automatically.

ANSWER BOUNDARY

Do not inspect device, software-user or network content; infer availability from power draw; grant a booking; or promise service. IT and room owners confirm readiness. The service cannot create, approve, publish or change the timetable.

WHY THIS IS USEFUL

Prevents a timetable from placing a practical in a room that cannot deliver the required computing environment.

TT-024

WHOLE-ROUTE ACCESSIBILITY

L3 - recommendation

R2 - integration-dependent

TIMETABLING TEAM QUESTION

Which eligible rooms meet the authorised functional requirements across the full route, entrance, lift or door status, seating and assistive provision?

SENSORS AND TELEMETRY

Use owner-issued lift, door and route-segment status for the complete named route. Proposed anonymous directional counts may supplement only the covered corridor, stair and lift-lobby boundaries, at 1-minute sampling aggregated over 5 minutes. Require commissioned mapping, synchronized clocks, eligible coverage, uptime/gap, drift, maintenance and relocation status; no identities, cameras, device tracking or linked movement histories.

AUTHORITATIVE SUPPORTING DATA

Current accessible-route and entrance records; owner-issued lift and door status; room seating and assistive provision; current room eligibility; authorised functional requirements supplied for scheduling; closures and access times. Source owners: central room-data and timetabling owners; room owner; IT/AV or specialist-equipment owner; accessibility service; Estates/Facilities for served assets. Record source version, effective/observed and retrieved times, permission scope, staleness and conflicts. Current owner-issued records prevail over history, inference or telemetry; equal-authority conflicts remain explicit.

ANALYSIS REQUIRED

Operation: whole-route validation and constrained recommendation. Decision window: room allocation and each affected session. Validate every route segment and destination provision, then list only eligible rooms. Missing segment or stale status produces a qualified alternative or not assessable. Exclude telemetry that fails mapping, clock, coverage or health checks; state excluded intervals and residual uncertainty. Filter hard constraints first, compare only eligible options under explicit criteria, show alternatives and never apply changes automatically.

ANSWER BOUNDARY

Do not request a diagnosis, disclose adjustment reasons, certify accessibility or safety, or assume apparent use proves suitability. Accessibility and asset owners retain authority. Do not substitute an uncovered route segment or adjacent boundary for complete route evidence. If mapping, timing, coverage or sensor health is inadequate, exclude the stream; do not impute or substitute nearby data. The service cannot create, approve, publish or change the timetable.

WHY THIS IS USEFUL

Lets the team avoid room allocations that would make an authorised activity inaccessible.

3. Room capacity, layout, equipment and accessibility suitability

TT-025 to TT-026 | 10 questions in this section | Filter by current room eligibility, whole-route access, equipment, service and evidence coverage before comparing preferences.

TT-025

HYBRID AND CAPTURE READINESS

L3 - recommendation

R2 - integration-dependent

TIMETABLING TEAM QUESTION

Which currently eligible rooms are approved for the required hybrid, lecture-capture or assistive-technology setup?

SENSORS AND TELEMETRY
Use only owner-issued non-content status for each named room or served asset: door/lift state, workstation or AV inventory/health, network/power availability, setup state or maintenance flag as relevant. Preserve asset-to-room mapping, event and retrieval times, heartbeat, maintenance mode, permission and conflicts. Do not inspect screens, files, traffic or teaching content.

AUTHORITATIVE SUPPORTING DATA
Current room AV and capture inventory; approved teaching mode; service health; network and power status; assistive-technology provision; room eligibility; maintenance and support windows. Source owners: central room-data and timetabling owners; room owner; IT/AV or specialist-equipment owner; accessibility service; Estates/Facilities for served assets. Record source version, effective/observed and retrieved times, permission scope, staleness and conflicts. Current owner-issued records prevail over history, inference or telemetry; equal-authority conflicts remain explicit.

ANALYSIS REQUIRED
Operation: authorised lookup plus suitability recommendation. Decision window: room allocation and pre-session recheck. Require every mandatory service and owner-issued current status; rank only fully eligible rooms. Stale or conflicting service evidence requires deferral to AV or IT. Exclude telemetry that fails mapping, clock, coverage or health checks; state excluded intervals and residual uncertainty. Filter hard constraints first, compare only eligible options under explicit criteria, show alternatives and never apply changes automatically.

ANSWER BOUNDARY
Do not inspect teaching content, recordings, screens or network traffic; enable recording; operate equipment; or treat a device list as proof of current readiness. Service owners confirm operation. The service cannot create, approve, publish or change the timetable.

WHY THIS IS USEFUL

Reduces late room changes caused by assigning a session to an unsuitable hybrid-teaching space.

TT-026

ROOM ENVIRONMENT EVIDENCE

L3 - comparison

R3 - research or restricted

TIMETABLING TEAM QUESTION

Among otherwise eligible rooms, which have enough room-level evidence to compare likely conditions for this cohort size and session length?

SENSORS AND TELEMETRY
Proposed whole-room anonymous doorway or ceiling counting, plus room-level temperature, relative humidity and direct CO2 sensing; optional numerical LAeq only where proportionate. Use 1-minute observations and 15-minute aggregates. Require representative occupied-zone placement, complete count coverage, calibration/commissioning, synchronized clocks, uptime/gap, drift and relocation status. Corridor data is not room evidence.

AUTHORITATIVE SUPPORTING DATA
Current eligible-room set; room volume and served HVAC-zone mapping; scheduled cohort range and duration; BMS operating status; maintenance; comparable historical sessions; and the institution's chosen operational criteria. Source owners: central room-data and timetabling owners; room owner; IT/AV or specialist-equipment owner; accessibility service; Estates/Facilities for served assets. Record source version, effective/observed and retrieved times, permission scope, staleness and conflicts. Current owner-issued records prevail over history, inference or telemetry; equal-authority conflicts remain explicit.

ANALYSIS REQUIRED
Operation: evidence sufficiency check, matched comparison and qualified forecast. Decision window: room selection. Use comparable occupancy, season and HVAC operation; report coverage, uncertainty and missingness. If room-level coverage is absent, return not assessable rather than using corridor data. Exclude telemetry that fails mapping, clock, coverage or health checks; state excluded intervals and residual uncertainty. Compare only eligible cases under matched programme, room, time and occupancy conditions; disclose residual confounders.

ANSWER BOUNDARY
Do not certify comfort, health, ventilation or safety, diagnose individuals or override a room owner. Environmental telemetry is secondary context after capacity, access and equipment eligibility. Do not infer individual attendance, absence, engagement, performance or reasons from aggregate use. If mapping, timing, coverage or sensor health is inadequate, exclude the stream; do not impute or substitute nearby data. The service cannot create, approve, publish or change the timetable.

WHY THIS IS USEFUL

Helps choose between otherwise suitable rooms while being honest about what environmental evidence can support.

3. Room capacity, layout, equipment and accessibility suitability

TT-027 to TT-028 | 10 questions in this section | Filter by current room eligibility, whole-route access, equipment, service and evidence coverage before comparing preferences.

TT-027

SPECIALIST PRACTICAL ROOM

L2 - lookup

R2 - integration-dependent

TIMETABLING TEAM QUESTION

Which specialist rooms meet the authorised equipment, supervision, access and safety conditions for this practical?

SENSORS AND TELEMETRY

Use only owner-issued non-content status for each named room or served asset: door/lift state, workstation or AV inventory/health, network/power availability, setup state or maintenance flag as relevant. Preserve asset-to-room mapping, event and retrieval times, heartbeat, maintenance mode, permission and conflicts. Do not inspect screens, files, traffic or teaching content.

AUTHORITATIVE SUPPORTING DATA

Current specialist-room inventory and purpose; required equipment and service status; authorised supervision and technical support; risk-assessment conditions; capacity; access; booking eligibility; closures; and safety-owner approval. Source owners: central room-data and timetabling owners; room owner; IT/AV or specialist-equipment owner; accessibility service; Estates/Facilities for served assets. Record source version, effective/observed and retrieved times, permission scope, staleness and conflicts. Current owner-issued records prevail over history, inference or telemetry; equal-authority conflicts remain explicit.

ANALYSIS REQUIRED

Operation: authorised suitability lookup. Decision window: practical-room allocation. Filter only on current owner-approved requirements and return missing evidence separately. If safety, equipment or supervision status is not current, the room is not assessable. Exclude telemetry that fails mapping, clock, coverage or health checks; state excluded intervals and residual uncertainty. If minimum evidence is missing, stale, conflicting or unauthorised, qualify the answer or return not assessable.

ANSWER BOUNDARY

Do not infer hazards or suitability from historic room labels, operate equipment, issue safety approval or grant access. Competent room, technical and safety owners decide eligibility. The service cannot create, approve, publish or change the timetable.

WHY THIS IS USEFUL

Keeps specialist teaching in rooms that are currently authorised for the actual activity.

TT-028

EQUIVALENT ROOM ALTERNATIVES

L3 - recommendation

R2 - integration-dependent

TIMETABLING TEAM QUESTION

If the preferred room is unavailable, which alternatives preserve every essential teaching, capacity, access and service requirement?

SENSORS AND TELEMETRY

Use only owner-issued non-content status for each named room or served asset: door/lift state, workstation or AV inventory/health, network/power availability, setup state or maintenance flag as relevant. Preserve asset-to-room mapping, event and retrieval times, heartbeat, maintenance mode, permission and conflicts. Do not inspect screens, files, traffic or teaching content.

AUTHORITATIVE SUPPORTING DATA

Current room eligibility and booking status; approved capacity and layout; required equipment, AV and network; accessible route and seating; setup time; location; and closure or maintenance status. Source owners: central room-data and timetabling owners; room owner; IT/AV or specialist-equipment owner; accessibility service; Estates/Facilities for served assets. Record source version, effective/observed and retrieved times, permission scope, staleness and conflicts. Current owner-issued records prevail over history, inference or telemetry; equal-authority conflicts remain explicit.

ANALYSIS REQUIRED

Operation: equivalence filtering followed by recommendation. Decision window: room replacement. Separate essential requirements from preferences, exclude any failed hard constraint, and compare transition and setup impacts. If no fully equivalent room exists, say so and show the missing requirement. Exclude telemetry that fails mapping, clock, coverage or health checks; state excluded intervals and residual uncertainty. Filter hard constraints first, compare only eligible options under explicit criteria, show alternatives and never apply changes automatically.

ANSWER BOUNDARY

Do not label a room equivalent because it is empty or nearby, grant a booking or relax essential requirements. Responsible owners approve any compromise. The service cannot create, approve, publish or change the timetable.

WHY THIS IS USEFUL

Provides a trustworthy fallback shortlist during routine or late room changes.

3. Room capacity, layout, equipment and accessibility suitability

TT-029 to TT-030 | 10 questions in this section | Filter by current room eligibility, whole-route access, equipment, service and evidence coverage before comparing preferences.

TT-029

ROOM DATA GAPS

L3 - diagnosis

R2 - integration-dependent

TIMETABLING TEAM QUESTION

Which rooms must be excluded because current capacity, layout, access, equipment, service or ownership evidence is incomplete or contradictory?

SENSORS AND TELEMETRY

Use the proposed evidence inventory and quality metadata rather than substituting observations: exact room/route/asset mapping, deployment and relocation history, commissioning/calibration, clock offset, last heartbeat, uptime, gaps, drift, maintenance and permission state. Where aggregate counts or environment streams are relevant, retain their documented error and eligible coverage.

AUTHORITATIVE SUPPORTING DATA

Current room inventory, ownership, capacity, layout, booking eligibility, accessibility, equipment, service and closure sources; source-version map; and conflict-resolution owner. Source owners: central room-data and timetabling owners; room owner; IT/AV or specialist-equipment owner; accessibility service; Estates/Facilities for served assets. Record source version, effective/observed and retrieved times, permission scope, staleness and conflicts. Current owner-issued records prevail over history, inference or telemetry; equal-authority conflicts remain explicit.

ANALYSIS REQUIRED

Operation: evidence-quality diagnosis. Decision window: room-inventory readiness. Check mandatory fields, currentness and cross-source agreement, then classify each room as assessable, conditionally assessable or not assessable. Do not silently impute missing values. Exclude telemetry that fails mapping, clock, coverage or health checks; state excluded intervals and residual uncertainty. Identify the minimum conflict set or temporal association; do not claim causation.

ANSWER BOUNDARY

Do not infer room purpose or capacity from the May 2021 plan, sensor location or occupancy. Data owners correct the authoritative record. It may report an association, not an unverified cause. The service cannot create, approve, publish or change the timetable.

WHY THIS IS USEFUL

Prevents incomplete room records from entering the candidate set as if they were verified.

TT-030

CHANGEOVER AND RESET TIME

L2 - calculation

R2 - integration-dependent

TIMETABLING TEAM QUESTION

Is there enough verified changeover time between consecutive activities when layouts, equipment or access arrangements differ?

SENSORS AND TELEMETRY

No physical sensing is needed for the core calculation. Proposed anonymous room-level doorway or ceiling counts may be used only as secondary historical evidence, with complete entrance coverage, commissioned error bounds, aligned clocks, 1-minute observations aggregated to 15 minutes, and uptime, gap, drift and relocation flags. Counts never replace enrolment, capacity or booking records.

AUTHORITATIVE SUPPORTING DATA

Current consecutive bookings; room layout and reset requirements; equipment setup; cleaning or technical turnaround where authorised; cohort size range; access arrangements; and owner-approved minimum interval. Source owners: central room-data and timetabling owners; room owner; IT/AV or specialist-equipment owner; accessibility service; Estates/Facilities for served assets. Record source version, effective/observed and retrieved times, permission scope, staleness and conflicts. Current owner-issued records prevail over history, inference or telemetry; equal-authority conflicts remain explicit.

ANALYSIS REQUIRED

Operation: turnaround calculation. Decision window: the gap between two bookings. Sum egress, reset, setup and entry allowances, then compare with the scheduled gap. Use observed aggregate delay only as qualified context. If a required component is unknown, return not assessable. Exclude telemetry that fails mapping, clock, coverage or health checks; state excluded intervals and residual uncertainty. If minimum evidence is missing, stale, conflicting or unauthorised, qualify the answer or return not assessable.

ANSWER BOUNDARY

Do not instruct operational teams, infer readiness from an empty room or change owner-defined turnaround standards. The service flags an infeasible gap for responsible review. The service cannot create, approve, publish or change the timetable.

WHY THIS IS USEFUL

Avoids back-to-back allocations that cannot be reset or accessed in time.

4. Clashes, travel time and campus-to-campus transitions

TT-031 to TT-032 | 10 questions in this section | Protect cohort and staff transitions using current routes, complete asset status and qualified aggregate flow evidence.

TT-031

REMAINING COHORT CLASHES

L2 - calculation

R1 - practical

TIMETABLING TEAM QUESTION

Which protected cohort clashes remain in the current draft, including partial overlaps?

SENSORS AND TELEMETRY

No physical sensing is needed for the core result. Where building-open, closure or service status matters, use only the current owner-issued feed for the named asset, with verified mapping, observed/retrieved time, heartbeat, maintenance state and conflicts. Telemetry does not create permission or availability.

AUTHORITATIVE SUPPORTING DATA

Current draft timetable; approved cohort and pathway clash matrix; session times; module instances; split-group rules; and exception approvals. Source owners: central timetabling owner; room and route owners; accessibility service; Estates/Facilities for lifts and doors; authorised transport source where relevant. Record source version, effective/observed and retrieved times, permission scope, staleness and conflicts. Current owner-issued records prevail over history, inference or telemetry; equal-authority conflicts remain explicit.

ANALYSIS REQUIRED

Operation: interval-overlap calculation. Decision window: draft review. Test all protected combinations, distinguish full and partial overlap and identify the owning protection rule. Missing cohort mapping is reported as incomplete coverage. Validate version, effective time, permission and conflicts; never infer hard constraints from history or telemetry. If minimum evidence is missing, stale, conflicting or unauthorised, qualify the answer or return not assessable.

ANSWER BOUNDARY

Do not expose individual enrolments, infer attendance or decide which session should move. The service locates conflicts; authorised timetablers and programme owners resolve them. The service cannot create, approve, publish or change the timetable.

WHY THIS IS USEFUL

Provides a complete clash list before publication or a major draft review.

TT-032

STAFF TRAVEL FEASIBILITY

L2 - calculation

R2 - integration-dependent

TIMETABLING TEAM QUESTION

Which back-to-back staff assignments leave less than the authorised travel and setup time between locations?

SENSORS AND TELEMETRY

Use owner-issued lift, door and route-segment status for the complete named route. Proposed anonymous directional counts may supplement only the covered corridor, stair and lift-lobby boundaries, at 1-minute sampling aggregated over 5 minutes. Require commissioned mapping, synchronized clocks, eligible coverage, uptime/gap, drift, maintenance and relocation status; no identities, cameras, device tracking or linked movement histories.

AUTHORITATIVE SUPPORTING DATA

Permission-bounded staff teaching assignments; current room locations; eligible route graph; lift, door and access status; authorised transition allowance; setup duties; and campus transport where relevant. Source owners: central timetabling owner; room and route owners; accessibility service; Estates/Facilities for lifts and doors; authorised transport source where relevant. Record source version, effective/observed and retrieved times, permission scope, staleness and conflicts. Current owner-issued records prevail over history, inference or telemetry; equal-authority conflicts remain explicit.

ANALYSIS REQUIRED

Operation: whole-route travel calculation. Decision window: each consecutive assignment. Calculate a conservative route time plus setup margin and flag negative slack. Optional aggregate congestion may widen the interval but never identify an individual. Missing route status makes the result conditional. Exclude telemetry that fails mapping, clock, coverage or health checks; state excluded intervals and residual uncertainty. If minimum evidence is missing, stale, conflicting or unauthorised, qualify the answer or return not assessable.

ANSWER BOUNDARY

Do not expose personal schedules beyond authorised timetabling need, track staff, infer performance or guarantee travel time. Staff and room owners confirm constraints. Do not substitute an uncovered route segment or adjacent boundary for complete route evidence. If mapping, timing, coverage or sensor health is inadequate, exclude the stream; do not impute or substitute nearby data. The service cannot create, approve, publish or change the timetable.

WHY THIS IS USEFUL

Finds transitions that are technically conflict-free but practically impossible.

4. Clashes, travel time and campus-to-campus transitions

TT-033 to TT-034 | 10 questions in this section | Protect cohort and staff transitions using current routes, complete asset status and qualified aggregate flow evidence.

TT-033

STEP-FREE TRANSITION TIME

L3 - recommendation

R2 - integration-dependent

TIMETABLING TEAM QUESTION

Can each consecutive session be reached within the gap by the authorised step-free route, and which alternatives preserve that requirement?

SENSORS AND TELEMETRY

Use owner-issued lift, door and route-segment status for the complete named route. Proposed anonymous directional counts may supplement only the covered corridor, stair and lift-lobby boundaries, at 1-minute sampling aggregated over 5 minutes. Require commissioned mapping, synchronized clocks, eligible coverage, uptime/gap, drift, maintenance and relocation status; no identities, cameras, device tracking or linked movement histories.

AUTHORITATIVE SUPPORTING DATA

Current accessible-route graph; entrance, door and lift status; time-dependent access restrictions; destination room eligibility; authorised functional requirement; session times; and contingency routes. Source owners: central timetabling owner; room and route owners; accessibility service; Estates/Facilities for lifts and doors; authorised transport source where relevant. Record source version, effective/observed and retrieved times, permission scope, staleness and conflicts. Current owner-issued records prevail over history, inference or telemetry; equal-authority conflicts remain explicit.

ANALYSIS REQUIRED

Operation: whole-route calculation and constrained recommendation. Decision window: each transition. Validate every segment, add conservative transfer and setup time, and compare eligible alternative rooms or times. If any segment status is stale, return a qualified route or not assessable. Exclude telemetry that fails mapping, clock, coverage or health checks; state excluded intervals and residual uncertainty. Filter hard constraints first, compare only eligible options under explicit criteria, show alternatives and never apply changes automatically.

ANSWER BOUNDARY

Do not infer disability, demand a diagnosis, certify route accessibility or operate doors or lifts. Use only the functional requirement authorised for scheduling; competent owners retain authority. Do not substitute an uncovered route segment or adjacent boundary for complete route evidence. If mapping, timing, coverage or sensor health is inadequate, exclude the stream; do not impute or substitute nearby data. The service cannot create, approve, publish or change the timetable.

WHY THIS IS USEFUL

Allows the timetable to preserve a usable transition rather than merely a nominal gap.

TT-034

CROSS-CAMPUS TRANSITIONS

L3 - estimate

R2 - integration-dependent

TIMETABLING TEAM QUESTION

Which consecutive sessions require cross-campus travel, and is the approved transition allowance realistic for the current route and service pattern?

SENSORS AND TELEMETRY

No physical sensing is needed for the core result. Proposed anonymous counts may qualify only a covered campus approach or building boundary, with commissioned location, aligned clocks, 5-minute aggregates, uptime and gap flags; they cannot represent an individual journey.

AUTHORITATIVE SUPPORTING DATA

Current session locations and times; campus route graph; public or university transport timetable and current service status; access restrictions; authorised transition allowance; and relevant weather or closure notices. Source owners: central timetabling owner; room and route owners; accessibility service; Estates/Facilities for lifts and doors; authorised transport source where relevant. Record source version, effective/observed and retrieved times, permission scope, staleness and conflicts. Current owner-issued records prevail over history, inference or telemetry; equal-authority conflicts remain explicit.

ANALYSIS REQUIRED

Operation: route-time estimate. Decision window: each cross-campus transition. Combine scheduled travel, walking and setup margins; use service uncertainty and a conservative buffer. If the transport status or route is not current, return a range or not assessable. Exclude telemetry that fails mapping, clock, coverage or health checks; state excluded intervals and residual uncertainty. Use eligible comparators, report ranges and sensitivity, and return not assessable when coverage or drift is inadequate.

ANSWER BOUNDARY

Do not track individuals, guarantee transport, infer attendance or alter employment or accessibility constraints. Transport and campus owners remain authoritative. Do not substitute an uncovered route segment or adjacent boundary for complete route evidence. The service cannot create, approve, publish or change the timetable.

WHY THIS IS USEFUL

Identifies cross-campus sequences that need a different time or location before they become recurring problems.

4. Clashes, travel time and campus-to-campus transitions

TT-035 to TT-036 | 10 questions in this section | Protect cohort and staff transitions using current routes, complete asset status and qualified aggregate flow evidence.

TT-035

ROUTE OR LIFT OUTAGE

L3 - recommendation

R2 - integration-dependent

TIMETABLING TEAM QUESTION

If a required lift, door or route segment is unavailable, which consecutive sessions become unreachable and which authorised alternatives remain?

SENSORS AND TELEMETRY

Use owner-issued lift, door and route-segment status for the complete named route. Proposed anonymous directional counts may supplement only the covered corridor, stair and lift-lobby boundaries, at 1-minute sampling aggregated over 5 minutes. Require commissioned mapping, synchronized clocks, eligible coverage, uptime/gap, drift, maintenance and relocation status; no identities, cameras, device tracking or linked movement histories.

AUTHORITATIVE SUPPORTING DATA

Owner-issued outage and restoration status; current accessible and general route graphs; affected session locations and times; room eligibility; available alternatives; and official closure or safety notices. Source owners: central timetabling owner; room and route owners; accessibility service; Estates/Facilities for lifts and doors; authorised transport source where relevant. Record source version, effective/observed and retrieved times, permission scope, staleness and conflicts. Current owner-issued records prevail over history, inference or telemetry; equal-authority conflicts remain explicit.

ANALYSIS REQUIRED

Operation: impact calculation and contingency recommendation. Decision window: the outage window. Remove unavailable segments, recompute eligible transitions and compare room or time alternatives. If restoration timing is uncertain, show time-bounded scenarios and recheck time. Exclude telemetry that fails mapping, clock, coverage or health checks; state excluded intervals and residual uncertainty. Filter hard constraints first, compare only eligible options under explicit criteria, show alternatives and never apply changes automatically.

ANSWER BOUNDARY

Do not reopen routes, operate assets, certify safety or replace official instructions. Telemetry never overrides facilities, security or emergency communications. Do not substitute an uncovered route segment or adjacent boundary for complete route evidence. If mapping, timing, coverage or sensor health is inadequate, exclude the stream; do not impute or substitute nearby data. The service cannot create, approve, publish or change the timetable.

WHY THIS IS USEFUL

Supports a timely room or time review when the original transition is no longer feasible.

TT-036

CLASS-CHANGE CONGESTION

L3 - forecast

R3 - research or restricted

TIMETABLING TEAM QUESTION

Where does the draft timetable risk unreliable class-change flow at covered corridor, stair or lift boundaries?

SENSORS AND TELEMETRY

Proposed anonymous directional counts at every relevant corridor, stair, lift-lobby or doorway boundary; 1-minute observations aggregated over 5 minutes. Require complete boundary coverage, commissioning against reference counts, synchronized clocks, uptime/gap, drift, maintenance and relocation status. No identities, cameras, device tracking or linked movement histories.

AUTHORITATIVE SUPPORTING DATA

Current draft timetable; room capacities and exits; verified route graph; class-change windows; owner-issued route or lift status; and comparable aggregate counts from covered boundaries. Source owners: central timetabling owner; room and route owners; accessibility service; Estates/Facilities for lifts and doors; authorised transport source where relevant. Record source version, effective/observed and retrieved times, permission scope, staleness and conflicts. Current owner-issued records prevail over history, inference or telemetry; equal-authority conflicts remain explicit.

ANALYSIS REQUIRED

Operation: aggregate flow forecast. Decision window: each 10- to 20-minute changeover. Estimate demand from the timetable, calibrate against comparable covered periods and report confidence, uncovered routes and sensitivity. If coverage is insufficient, provide timetable demand only, not observed congestion. Exclude telemetry that fails mapping, clock, coverage or health checks; state excluded intervals and residual uncertainty. Use eligible comparators, report ranges and sensitivity, and return not assessable when coverage or drift is inadequate.

ANSWER BOUNDARY

Do not identify or track people, reconstruct paths, certify evacuation or safety, or treat counts as exact attendance. Route and safety owners decide operational action. Uncovered route segments or failed sensor-quality checks make the flow estimate qualified or not assessable. The service cannot create, approve, publish or change the timetable.

WHY THIS IS USEFUL

Enables start-time, room or transition-buffer changes before recurrent bottlenecks are created.

4. Clashes, travel time and campus-to-campus transitions

TT-037 to TT-038 | 10 questions in this section | Protect cohort and staff transitions using current routes, complete asset status and qualified aggregate flow evidence.

TT-037

USABLE MEAL AND REST BREAKS

L2 - calculation

R1 - practical

TIMETABLING TEAM QUESTION

Which full-day cohort patterns lack an authorised usable meal or rest break after realistic transitions are removed?

SENSORS AND TELEMETRY

No physical sensing is needed for the core result. Where building-open, closure or service status matters, use only the current owner-issued feed for the named asset, with verified mapping, observed/retrieved time, heartbeat, maintenance state and conflicts. Telemetry does not create permission or availability.

AUTHORITATIVE SUPPORTING DATA

Current draft timetable; cohort schedule; protected-break policy; verified room locations; transition allowances; teaching-day limits; and authorised exceptions. Source owners: central timetabling owner; room and route owners; accessibility service; Estates/Facilities for lifts and doors; authorised transport source where relevant. Record source version, effective/observed and retrieved times, permission scope, staleness and conflicts. Current owner-issued records prevail over history, inference or telemetry; equal-authority conflicts remain explicit.

ANALYSIS REQUIRED

Operation: schedule-window calculation. Decision window: each cohort day. Subtract taught and transition periods, then test remaining continuous gaps against policy. If route times are uncertain, report the break as a range and flag borderline cases. Validate version, effective time, permission and conflicts; never infer hard constraints from history or telemetry. If minimum evidence is missing, stale, conflicting or unauthorised, qualify the answer or return not assessable.

ANSWER BOUNDARY

Do not infer an individual's dietary, health or attendance needs, or redefine the protected-break rule. Programme and policy owners decide exceptions. The service cannot create, approve, publish or change the timetable.

WHY THIS IS USEFUL

Reveals days that look compliant by clock time but leave no practically usable break.

TT-038

OPTION COMBINATION CLASHES

L3 - comparison

R2 - integration-dependent

TIMETABLING TEAM QUESTION

Do common authorised option combinations create hidden clashes or travel sequences that cannot be completed?

SENSORS AND TELEMETRY

No physical sensing is needed for the core result. Where building-open, closure or service status matters, use only the current owner-issued feed for the named asset, with verified mapping, observed/retrieved time, heartbeat, maintenance state and conflicts. Telemetry does not create permission or availability.

AUTHORITATIVE SUPPORTING DATA

Authorised aggregate option combinations; draft timetable; room locations; verified route graph; transition policy; suppression rules for small groups; and current access status. Source owners: central timetabling owner; room and route owners; accessibility service; Estates/Facilities for lifts and doors; authorised transport source where relevant. Record source version, effective/observed and retrieved times, permission scope, staleness and conflicts. Current owner-issued records prevail over history, inference or telemetry; equal-authority conflicts remain explicit.

ANALYSIS REQUIRED

Operation: combination-based clash and route comparison. Decision window: draft review. Test eligible combinations, distinguish exact overlap from travel risk and state population coverage. Missing or disclosure-sensitive combinations are excluded and labelled. Validate version, effective time, permission and conflicts; never infer hard constraints from history or telemetry. Compare only eligible cases under matched programme, room, time and occupancy conditions; disclose residual confounders.

ANSWER BOUNDARY

Do not expose or predict individual choices, rank modules or decide which combinations students should take. Programme owners set clash-protection priorities. The service cannot create, approve, publish or change the timetable.

WHY THIS IS USEFUL

Finds conflicts that a compulsory-cohort check cannot see.

4. Clashes, travel time and campus-to-campus transitions

TT-039 to TT-040 | 10 questions in this section | Protect cohort and staff transitions using current routes, complete asset status and qualified aggregate flow evidence.

TT-039

LAST-MINUTE ROOM MOVE

L3 - recommendation

R2 - integration-dependent

TIMETABLING TEAM QUESTION

Would this proposed room move create an infeasible transition for the cohort, teaching staff or an authorised step-free route?

SENSORS AND TELEMETRY

Use owner-issued lift, door and route-segment status for the complete named route. Proposed anonymous directional counts may supplement only the covered corridor, stair and lift-lobby boundaries, at 1-minute sampling aggregated over 5 minutes. Require commissioned mapping, synchronized clocks, eligible coverage, uptime/gap, drift, maintenance and relocation status; no identities, cameras, device tracking or linked movement histories.

AUTHORITATIVE SUPPORTING DATA

Confirmed proposed move; current and replacement room eligibility; cohort and staff adjacent sessions; route graph; lift and door status; setup time; access restrictions; and notification deadline. Source owners: central timetabling owner; room and route owners; accessibility service; Estates/Facilities for lifts and doors; authorised transport source where relevant. Record source version, effective/observed and retrieved times, permission scope, staleness and conflicts. Current owner-issued records prevail over history, inference or telemetry; equal-authority conflicts remain explicit.

ANALYSIS REQUIRED

Operation: change-impact calculation and recommendation. Decision window: the affected day. Recompute every adjacent transition and room requirement, show negative margins and list only fully eligible alternatives. If route or room status is stale, do not recommend the move. Exclude telemetry that fails mapping, clock, coverage or health checks; state excluded intervals and residual uncertainty. Filter hard constraints first, compare only eligible options under explicit criteria, show alternatives and never apply changes automatically.

ANSWER BOUNDARY

Do not grant the room, change the timetable, track individuals or certify accessibility. Responsible room, accessibility and timetable owners approve the change. Do not substitute an uncovered route segment or adjacent boundary for complete route evidence. If mapping, timing, coverage or sensor health is inadequate, exclude the stream; do not impute or substitute nearby data. The service cannot create, approve, publish or change the timetable.

WHY THIS IS USEFUL

Prevents a local room fix from creating a new travel or access failure elsewhere.

TT-040

REDUCING AVOIDABLE TRAVEL

L3 - recommendation

R3 - research or restricted

TIMETABLING TEAM QUESTION

Which timetable changes could reduce avoidable campus travel while preserving every academic, employment, accessibility and room constraint?

SENSORS AND TELEMETRY

No physical sensing is needed for the core schedule comparison. Proposed anonymous directional counters may qualify recurrent congestion only at the exact corridor, stair or lift-lobby boundaries used by the route, using 1-minute observations and 5-minute aggregates. Require commissioned coverage, synchronized clocks, uptime/gap, drift and relocation status; no identities or path histories.

AUTHORITATIVE SUPPORTING DATA

Current draft; verified room locations and routes; cohort and staff transition constraints; accessibility requirements; room eligibility; programme sequence rules; and explicit optimisation criteria. Source owners: central timetabling owner; room and route owners; accessibility service; Estates/Facilities for lifts and doors; authorised transport source where relevant. Record source version, effective/observed and retrieved times, permission scope, staleness and conflicts. Current owner-issued records prevail over history, inference or telemetry; equal-authority conflicts remain explicit.

ANALYSIS REQUIRED

Operation: constrained scenario optimisation and recommendation. Decision window: draft redesign. Minimise aggregate transition distance or time subject to all hard constraints, compare trade-offs and show sensitivity to uncertain routes. If the objective or constraints are incomplete, provide comparison only. Exclude telemetry that fails mapping, clock, coverage or health checks; state excluded intervals and residual uncertainty. Filter hard constraints first, compare only eligible options under explicit criteria, show alternatives and never apply changes automatically.

ANSWER BOUNDARY

Do not rank individuals, infer travel preferences or override protected constraints for a shorter route. Authorised owners decide whether to adopt a scenario. Do not substitute an uncovered route segment or adjacent boundary for complete route evidence. The service cannot create, approve, publish or change the timetable.

WHY THIS IS USEFUL

Supports a more navigable timetable without trading away teaching or accessibility requirements.

5. Occupancy, attendance and timetable-feedback evidence

TT-041 to TT-042 | 10 questions in this section | Use anonymous aggregate evidence to review scheduling patterns without inferring who attended, why, or how anyone performed.

TT-041

ROOM RIGHT-SIZING REVIEW

L3 - comparison

R2 - integration-dependent

TIMETABLING TEAM QUESTION

Which recurring sessions appear consistently over-roomed when authorised cohort need and reliable anonymous use are considered together?

SENSORS AND TELEMETRY

Proposed anonymous whole-room counts from reconciled doorway counters or privacy-preserving ceiling sensing; 1-minute observations aggregated to 15 minutes. Require every entrance to be covered, commissioned error bounds, synchronized clocks, uptime/gap, drift, maintenance and relocation status. Counts are room-use evidence only - never identities, attendance, engagement or authorised capacity.

AUTHORITATIVE SUPPORTING DATA

Current timetable and room booking; authorised room capacity by layout; cohort or enrolment snapshot; approved activity requirements; exceptional-session tags; and comparable aggregate count periods. Source owners: central timetabling and room owners; student-records owner for aggregate cohort data; sensor-data steward; authorised feedback owner. Record source version, effective/observed and retrieved times, permission scope, staleness and conflicts. Current owner-issued records prevail over history, inference or telemetry; equal-authority conflicts remain explicit.

ANALYSIS REQUIRED

Operation: matched comparison. Decision window: next-cycle room review. Compare scheduled cohort range and quality-controlled aggregate use across comparable occurrences, calculate sustained capacity margin and report coverage and uncertainty. Missing or unreliable counts cannot support downsizing. Exclude telemetry that fails mapping, clock, coverage or health checks; state excluded intervals and residual uncertainty. Compare only eligible cases under matched programme, room, time and occupancy conditions; disclose residual confounders.

ANSWER BOUNDARY

Do not identify attendees, infer individual attendance or performance, treat a low count as permission or change authorised capacity. Room and programme owners approve any move. If mapping, timing, coverage or sensor health is inadequate, exclude the stream; do not impute or substitute nearby data. The service cannot create, approve, publish or change the timetable.

WHY THIS IS USEFUL

Identifies credible candidates for a better-sized room without relying on one low-attendance session.

TT-042

NEAR-CAPACITY USE

L3 - forecast

R2 - integration-dependent

TIMETABLING TEAM QUESTION

Which sessions repeatedly operate near authorised room capacity and may need a larger room or different group structure?

SENSORS AND TELEMETRY

Proposed anonymous whole-room counts from reconciled doorway counters or privacy-preserving ceiling sensing; 1-minute observations aggregated to 15 minutes. Require every entrance to be covered, commissioned error bounds, synchronized clocks, uptime/gap, drift, maintenance and relocation status. Counts are room-use evidence only - never identities, attendance, engagement or authorised capacity.

AUTHORITATIVE SUPPORTING DATA

Current room capacity by layout; timetable; cohort snapshot; group structure; comparable anonymous aggregate counts; exceptional-event tags; room access and booking status; and capacity owner. Source owners: central timetabling and room owners; student-records owner for aggregate cohort data; sensor-data steward; authorised feedback owner. Record source version, effective/observed and retrieved times, permission scope, staleness and conflicts. Current owner-issued records prevail over history, inference or telemetry; equal-authority conflicts remain explicit.

ANALYSIS REQUIRED

Operation: threshold calculation plus next-cycle forecast. Decision window: each recurring series. Estimate count error, calculate frequency and duration near capacity and test the authorised enrolment range. Report a range and exclude weeks with incomplete coverage. Exclude telemetry that fails mapping, clock, coverage or health checks; state excluded intervals and residual uncertainty. Use eligible comparators, report ranges and sensitivity, and return not assessable when coverage or drift is inadequate.

ANSWER BOUNDARY

Do not treat counts as exact attendance, certify safety or automatically increase group size or room capacity. If coverage or capacity is unverified, return not assessable. If mapping, timing, coverage or sensor health is inadequate, exclude the stream; do not impute or substitute nearby data. The service cannot create, approve, publish or change the timetable.

WHY THIS IS USEFUL

Highlights fragile allocations before modest enrolment growth makes them infeasible.

5. Occupancy, attendance and timetable-feedback evidence

TT-043 to TT-044 | 10 questions in this section | Use anonymous aggregate evidence to review scheduling patterns without inferring who attended, why, or how anyone performed.

TT-043

APPARENT NON-USE

L3 - diagnosis

R2 - integration-dependent

TIMETABLING TEAM QUESTION

Which scheduled sessions repeatedly show no reliable anonymous room-use signal, and what evidence is needed before owner review?

SENSORS AND TELEMETRY
Proposed anonymous whole-room counts from reconciled doorway counters or privacy-preserving ceiling sensing; 1-minute observations aggregated to 15 minutes. Require every entrance to be covered, commissioned error bounds, synchronized clocks, uptime/gap, drift, maintenance and relocation status. Counts are room-use evidence only - never identities, attendance, engagement or authorised capacity.

AUTHORITATIVE SUPPORTING DATA
Current timetable and room booking; activity cancellation or relocation records; room access and closure status; sensor health; authorised aggregate use evidence; and session-owner contact route. Source owners: central timetabling and room owners; student-records owner for aggregate cohort data; sensor-data steward; authorised feedback owner. Record source version, effective/observed and retrieved times, permission scope, staleness and conflicts. Current owner-issued records prevail over history, inference or telemetry; equal-authority conflicts remain explicit.

ANALYSIS REQUIRED
Operation: evidence diagnosis. Decision window: several comparable occurrences. Require a healthy covered sensor, valid booking and no authorised cancellation; classify as supported non-use, conflicting or not assessable. A single missing count is never enough. Exclude telemetry that fails mapping, clock, coverage or health checks; state excluded intervals and residual uncertainty. Identify the minimum conflict set or temporal association; do not claim causation.

ANSWER BOUNDARY
Do not infer who attended, why a session was unused, teaching quality or productivity, and do not cancel or release a room. The owner reviews the evidence. Failed mapping, timing, coverage or sensor-health checks require a qualified or not-assessable answer. The service cannot create, approve, publish or change the timetable.

WHY THIS IS USEFUL

Provides a safe way to identify possible stale bookings without treating sensor silence as fact.

TT-044

OCCUPANCY EVIDENCE QUALITY

L2 - calculation

R2 - integration-dependent

TIMETABLING TEAM QUESTION

Is the anonymous occupancy evidence for this room and teaching period reliable enough to support timetable review?

SENSORS AND TELEMETRY
Use the proposed evidence inventory and quality metadata rather than substituting observations: exact room/route/asset mapping, deployment and relocation history, commissioning/calibration, clock offset, last heartbeat, uptime, gaps, drift, maintenance and permission state. Where aggregate counts or environment streams are relevant, retain their documented error and eligible coverage.

AUTHORITATIVE SUPPORTING DATA
Current sensor registry and room mapping; commissioning and calibration record; maintenance log; known outages; timetable interval; room closure or access status; and data-quality acceptance rule. Source owners: central timetabling and room owners; student-records owner for aggregate cohort data; sensor-data steward; authorised feedback owner. Record source version, effective/observed and retrieved times, permission scope, staleness and conflicts. Current owner-issued records prevail over history, inference or telemetry; equal-authority conflicts remain explicit.

ANALYSIS REQUIRED
Operation: data-quality calculation and answerability assessment. Decision window: the requested comparison period. Compute eligible coverage, missingness, clock offset and unresolved anomalies; return pass, qualified or not assessable with reasons. Exclude telemetry that fails mapping, clock, coverage or health checks; state excluded intervals and residual uncertainty. If minimum evidence is missing, stale, conflicting or unauthorised, qualify the answer or return not assessable.

ANSWER BOUNDARY
Do not repair data by inventing values, infer attendance from CO2 or unrelated sensors, or use corridor coverage for the room. Data owners decide remediation. The service cannot create, approve, publish or change the timetable.

WHY THIS IS USEFUL

Stops weak telemetry from driving a room or scheduling decision.

5. Occupancy, attendance and timetable-feedback evidence

TT-045 to TT-046 | 10 questions in this section | Use anonymous aggregate evidence to review scheduling patterns without inferring who attended, why, or how anyone performed.

TT-045

NEXT-CYCLE ROOM SIZE

L3 - recommendation

R3 - research or restricted

TIMETABLING TEAM QUESTION

Does authorised cohort evidence and reliable aggregate use support reviewing this recurring session for a smaller or larger room next year?

SENSORS AND TELEMETRY

Proposed anonymous whole-room counts from reconciled doorway counters or privacy-preserving ceiling sensing; 1-minute observations aggregated to 15 minutes. Require every entrance to be covered, commissioned error bounds, synchronized clocks, uptime/gap, drift, maintenance and relocation status. Counts are room-use evidence only - never identities, attendance, engagement or authorised capacity.

AUTHORITATIVE SUPPORTING DATA

Current and planning enrolment ranges; approved room capacity and activity layout; timetable series; comparable aggregate use; curriculum or delivery changes; exception weeks; and room eligibility for the next cycle. Source owners: central timetabling and room owners; student-records owner for aggregate cohort data; sensor-data steward; authorised feedback owner. Record source version, effective/observed and retrieved times, permission scope, staleness and conflicts. Current owner-issued records prevail over history, inference or telemetry; equal-authority conflicts remain explicit.

ANALYSIS REQUIRED

Operation: evidence synthesis and recommendation. Decision window: next timetable cycle. Combine enrolment range, group rules and sustained aggregate use, test both smaller and larger eligible rooms and report uncertainty and exceptional weeks. If evidence conflicts, recommend review rather than a size. Exclude telemetry that fails mapping, clock, coverage or health checks; state excluded intervals and residual uncertainty. Filter hard constraints first, compare only eligible options under explicit criteria, show alternatives and never apply changes automatically.

ANSWER BOUNDARY

Do not forecast named attendance, infer engagement, grant a room or change group policy. Programme and room owners decide allocation. If mapping, timing, coverage or sensor health is inadequate, exclude the stream; do not impute or substitute nearby data. The service cannot create, approve, publish or change the timetable.

WHY THIS IS USEFUL

Supports measured room-size review while protecting against overreaction to short-term attendance variation.

TT-046

RECURRING CHANGEOVER PRESSURE

L3 - diagnosis

R2 - integration-dependent

TIMETABLING TEAM QUESTION

Which class-change windows repeatedly show aggregate congestion near teaching rooms, and which timetable features coincide with it?

SENSORS AND TELEMETRY

Proposed anonymous directional counts at every relevant corridor, stair, lift-lobby or doorway boundary; 1-minute observations aggregated over 5 minutes. Require complete boundary coverage, commissioning against reference counts, synchronized clocks, uptime/gap, drift, maintenance and relocation status. No identities, cameras, device tracking or linked movement histories.

AUTHORITATIVE SUPPORTING DATA

Current timetable; room capacities and exits; covered route boundaries; owner-issued route or lift status; comparable teaching weeks; event and closure tags; and current route graph. Source owners: central timetabling and room owners; student-records owner for aggregate cohort data; sensor-data steward; authorised feedback owner. Record source version, effective/observed and retrieved times, permission scope, staleness and conflicts. Current owner-issued records prevail over history, inference or telemetry; equal-authority conflicts remain explicit.

ANALYSIS REQUIRED

Operation: matched pattern diagnosis. Decision window: recurring changeovers. Align counts to timetable demand, compare equivalent weekdays and exclude outages or events; report association, coverage and uncertainty rather than causation. Uncovered routes remain not assessable. Exclude telemetry that fails mapping, clock, coverage or health checks; state excluded intervals and residual uncertainty. Identify the minimum conflict set or temporal association; do not claim causation.

ANSWER BOUNDARY

Do not identify or track people, infer individual attendance or issue crowd-control instructions. Telemetry may support timetable review only. Uncovered route segments or failed mapping, timing, coverage or health checks make the diagnosis qualified or not assessable. Report associations, not unverified causes; the service cannot create, approve, publish or change the timetable.

WHY THIS IS USEFUL

Shows where staggered times, different rooms or larger transition gaps may be worth testing.

5. Occupancy, attendance and timetable-feedback evidence

TT-047 to TT-048 | 10 questions in this section | Use anonymous aggregate evidence to review scheduling patterns without inferring who attended, why, or how anyone performed.

TT-047

TIMETABLE FEEDBACK CORROBORATION

L3 - comparison

R2 - integration-dependent

TIMETABLING TEAM QUESTION Which coded timetable concerns are corroborated by current room, timing or route evidence, and which remain unclear?

SENSORS AND TELEMETRY

Use only the minimum proposed aggregate room, route or environmental telemetry named by the coded concern, with exact spatial mapping, aligned clocks, eligible observation window, calibration/commissioning, uptime, gaps, drift and provenance. Do not inspect free-text submissions, respondent identity, teaching content or device/network data.

AUTHORITATIVE SUPPORTING DATA

Authorised aggregate feedback category and period; current timetable; room and route records; service or closure status; relevant quality-controlled telemetry; and responsible owner. Source owners: central timetabling and room owners; student-records owner for aggregate cohort data; sensor-data steward; authorised feedback owner. Record source version, effective/observed and retrieved times, permission scope, staleness and conflicts. Current owner-issued records prevail over history, inference or telemetry; equal-authority conflicts remain explicit.

ANALYSIS REQUIRED

Operation: evidence comparison. Decision window: the reported series or period. Translate the coded issue into an evidence obligation, compare only aligned sources and classify supported, conflicting, unsupported or not assessable. Missing context triggers a minimal clarification request. Exclude telemetry that fails mapping, clock, coverage or health checks; state excluded intervals and residual uncertainty. Compare only eligible cases under matched programme, room, time and occupancy conditions; disclose residual confounders.

ANSWER BOUNDARY

Do not disclose personal feedback, infer complainant identity or motive, rate teaching or staff, or treat absence of telemetry as proof the concern is invalid. Do not infer individual attendance, absence, engagement, performance or reasons from aggregate use. If mapping, timing, coverage or sensor health is inadequate, exclude the stream; do not impute or substitute nearby data. The service cannot create, approve, publish or change the timetable.

WHY THIS IS USEFUL

Helps the team decide which concerns warrant a timetable investigation and which need more context.

TT-048

TIME-OF-DAY USE PATTERNS

L3 - comparison

R3 - research or restricted

TIMETABLING TEAM QUESTION Do early-morning or late-afternoon sessions show a sustained difference in reliable aggregate room use compared with equivalent daytime sessions?

SENSORS AND TELEMETRY

Proposed anonymous whole-room counts from reconciled doorway counters or privacy-preserving ceiling sensing; 1-minute observations aggregated to 15 minutes. Require every entrance to be covered, commissioned error bounds, synchronized clocks, uptime/gap, drift, maintenance and relocation status. Counts are room-use evidence only - never identities, attendance, engagement or authorised capacity.

AUTHORITATIVE SUPPORTING DATA

Comparable recurring sessions; current timetable; authorised cohort size; room capacity and type; activity type; teaching week; cancellations; weather or disruption tags; and quality-controlled aggregate counts. Source owners: central timetabling and room owners; student-records owner for aggregate cohort data; sensor-data steward; authorised feedback owner. Record source version, effective/observed and retrieved times, permission scope, staleness and conflicts. Current owner-issued records prevail over history, inference or telemetry; equal-authority conflicts remain explicit.

ANALYSIS REQUIRED

Operation: matched comparison. Decision window: a defined term or year. Match by module or activity, room type, cohort range and teaching week, estimate count uncertainty and report effect range. If comparable cases are too few, return not assessable. Exclude telemetry that fails mapping, clock, coverage or health checks; state excluded intervals and residual uncertainty. Compare only eligible cases under matched programme, room, time and occupancy conditions; disclose residual confounders.

ANSWER BOUNDARY

Do not infer individual attendance, motivation, learning or staff performance, and do not use correlation alone to mandate a time change. Academic owners interpret the result. If mapping, timing, coverage or sensor health is inadequate, exclude the stream; do not impute or substitute nearby data. The service cannot create, approve, publish or change the timetable.

WHY THIS IS USEFUL

Provides evidence for reviewing unpopular time bands without reducing the issue to anecdote.

5. Occupancy, attendance and timetable-feedback evidence

TT-049 to TT-050 | 10 questions in this section | Use anonymous aggregate evidence to review scheduling patterns without inferring who attended, why, or how anyone performed.

TT-049

EXCEPTION-WEEK TAGGING

L2 - calculation

R2 - integration-dependent

TIMETABLING TEAM QUESTION

Which weeks should be excluded or analysed separately when comparing aggregate room use or timetable feedback?

SENSORS AND TELEMETRY

No new sensing is required. Existing proposed streams must retain raw timestamps and interval-level commissioning, calibration, clock-offset, outage, gap, drift, maintenance and relocation flags so exceptional periods can be excluded transparently without deleting or inventing observations.

AUTHORITATIVE SUPPORTING DATA

Current academic calendar; assessment and reading weeks; holidays; cancellations; industrial action or authorised disruption; room closures; unusual events; sensor outages; and module delivery changes. Source owners: central timetabling and room owners; student-records owner for aggregate cohort data; sensor-data steward; authorised feedback owner. Record source version, effective/observed and retrieved times, permission scope, staleness and conflicts. Current owner-issued records prevail over history, inference or telemetry; equal-authority conflicts remain explicit.

ANALYSIS REQUIRED

Operation: authorised lookup plus data-preparation classification. Decision window: the comparison dataset. Tag each interval with the owning exception, retain rather than delete it, and produce primary and sensitivity analyses. Unknown exceptions remain visible. Exclude telemetry that fails mapping, clock, coverage or health checks; state excluded intervals and residual uncertainty. If minimum evidence is missing, stale, conflicting or unauthorised, qualify the answer or return not assessable.

ANSWER BOUNDARY

Do not remove inconvenient data silently, infer reasons from occupancy patterns or label a week exceptional without an authoritative source. The analysis must disclose exclusions. The service cannot create, approve, publish or change the timetable.

WHY THIS IS USEFUL

Makes later timetable comparisons reproducible and less vulnerable to distorted exceptional weeks.

TT-050

EVIDENCE THRESHOLD FOR CHANGE

L3 - answerability

R3 - research or restricted

TIMETABLING TEAM QUESTION

Which timetable-feedback signals meet the agreed threshold for owner review, and which still need more evidence?

SENSORS AND TELEMETRY

Use the proposed evidence inventory and quality metadata rather than substituting observations: exact room/route/asset mapping, deployment and relocation history, commissioning/calibration, clock offset, last heartbeat, uptime, gaps, drift, maintenance and permission state. Where aggregate counts or environment streams are relevant, retain their documented error and eligible coverage.

AUTHORITATIVE SUPPORTING DATA

Current timetable and room data; pre-agreed review criteria; authorised aggregate feedback category; comparable periods; relevant telemetry; source conflicts; and responsible decision owner. Source owners: central timetabling and room owners; student-records owner for aggregate cohort data; sensor-data steward; authorised feedback owner. Record source version, effective/observed and retrieved times, permission scope, staleness and conflicts. Current owner-issued records prevail over history, inference or telemetry; equal-authority conflicts remain explicit.

ANALYSIS REQUIRED

Operation: evidence-sufficiency assessment. Decision window: the reported issue. Test minimum coverage, recurrence, source agreement and plausible confounders; return supported for review, conflicting, insufficient or not assessable. The threshold must be documented before use. Exclude telemetry that fails mapping, clock, coverage or health checks; state excluded intervals and residual uncertainty. Classify the result as assessable, qualified or not assessable against predeclared thresholds.

ANSWER BOUNDARY

Do not automate a timetable change, infer individuals or claim causality from correlation. Evidence may trigger review; authorised owners decide action. Do not infer individual attendance, absence, engagement, performance or reasons from aggregate use. The service cannot create, approve, publish or change the timetable.

WHY THIS IS USEFUL

Creates a consistent standard for escalating evidence without pretending every signal is conclusive.

6. Late changes, disruption and communication

TT-051 to TT-052 | 10 questions in this section | Propagate owner-confirmed changes, compare authorised alternatives and expose cross-system conflicts without operating services.

TT-051

ROOM CLOSURE CONTINGENCY

L3 - recommendation

R2 - integration-dependent

TIMETABLING TEAM QUESTION

If a teaching room closes tomorrow, which alternatives preserve the session's essential requirements with the least disruption?

SENSORS AND TELEMETRY

Use only owner-issued non-content status for each named room or served asset: door/lift state, workstation or AV inventory/health, network/power availability, setup state or maintenance flag as relevant. Preserve asset-to-room mapping, event and retrieval times, heartbeat, maintenance mode, permission and conflicts. Do not inspect screens, files, traffic or teaching content.

AUTHORITATIVE SUPPORTING DATA

Owner-issued closure and expected review time; affected booking; current eligible rooms; capacity, layout, access, equipment and service requirements; adjacent sessions; and notification deadline. Source owners: central timetabling and room owners; programme/HR owner where relevant; IT/AV service; Estates/Facilities; Security and official communications owners. Record source version, effective/observed and retrieved times, permission scope, staleness and conflicts. Current owner-issued records prevail over history, inference or telemetry; equal-authority conflicts remain explicit.

ANALYSIS REQUIRED

Operation: impact calculation and constrained recommendation. Decision window: the closure window. Remove the room, test only fully eligible alternatives and compare clash, transition and communication impact. If no equivalent option exists, state that explicitly and defer. Exclude telemetry that fails mapping, clock, coverage or health checks; state excluded intervals and residual uncertainty. Filter hard constraints first, compare only eligible options under explicit criteria, show alternatives and never apply changes automatically.

ANSWER BOUNDARY

Do not reopen the room, grant a booking, relax safety or accessibility requirements, or issue an official change. Room and timetable owners approve and publish the response. The service cannot create, approve, publish or change the timetable.

WHY THIS IS USEFUL

Gives the team a defensible replacement shortlist during a time-critical closure.

TT-052

DECLARED STAFF UNAVAILABILITY

L3 - calculation

R3 - research or restricted

TIMETABLING TEAM QUESTION

When authorised staff unavailability is reported, which affected sessions have feasible alternatives with the least knock-on disruption?

SENSORS AND TELEMETRY

No physical sensing is needed for the core result. Where building-open, closure or service status matters, use only the current owner-issued feed for the named asset, with verified mapping, observed/retrieved time, heartbeat, maintenance state and conflicts. Telemetry does not create permission or availability.

AUTHORITATIVE SUPPORTING DATA

Permission-bounded unavailable window; confirmed teaching assignment; current timetable; co-teaching or cover rules; cohort commitments; eligible rooms; academic constraints; and publication deadlines. Source owners: central timetabling and room owners; programme/HR owner where relevant; IT/AV service; Estates/Facilities; Security and official communications owners. Record source version, effective/observed and retrieved times, permission scope, staleness and conflicts. Current owner-issued records prevail over history, inference or telemetry; equal-authority conflicts remain explicit.

ANALYSIS REQUIRED

Operation: change-impact calculation followed by recommendation. Decision window: the unavailable period. Remove the affected assignment, test authorised alternatives and quantify moved sessions, new clashes and communications. If cover authority is absent, do not propose a person. Validate version, effective time, permission and conflicts; never infer hard constraints from history or telemetry. If minimum evidence is missing, stale, conflicting or unauthorised, qualify the answer or return not assessable.

ANSWER BOUNDARY

Do not disclose reasons, inspect personal calendars, rank staff, assign cover or change employment constraints. Responsible staffing and academic owners decide the solution. The service cannot create, approve, publish or change the timetable.

WHY THIS IS USEFUL

Helps the team understand the smallest feasible timetable response while protecting staff privacy.

6. Late changes, disruption and communication

TT-053 to TT-054 | 10 questions in this section | Propagate owner-confirmed changes, compare authorised alternatives and expose cross-system conflicts without operating services.

TT-053

LATE COHORT-SIZE CHANGE

L3 - recommendation

R2 - integration-dependent

TIMETABLING TEAM QUESTION

If the authorised cohort estimate changes late, which rooms or group structures cross their current limits first?

SENSORS AND TELEMETRY

No physical sensing is needed for the core calculation. Proposed anonymous room-level doorway or ceiling counts may be used only as secondary historical evidence, with complete entrance coverage, commissioned error bounds, aligned clocks, 1-minute observations aggregated to 15 minutes, and uptime, gap, drift and relocation flags. Counts never replace enrolment, capacity or booking records.

AUTHORITATIVE SUPPORTING DATA

Updated cohort snapshot and effective time; room capacities by layout; group-size rules; current room allocations; eligible alternatives; equipment and accessibility requirements; and change deadlines. Source owners: central timetabling and room owners; programme/HR owner where relevant; IT/AV service; Estates/Facilities; Security and official communications owners. Record source version, effective/observed and retrieved times, permission scope, staleness and conflicts. Current owner-issued records prevail over history, inference or telemetry; equal-authority conflicts remain explicit.

ANALYSIS REQUIRED

Operation: threshold calculation and contingency recommendation. Decision window: affected teaching series. Recalculate capacity margin and group count, identify the first breached constraint and list only authorised alternatives. If capacity or enrolment is provisional, report ranges. Exclude telemetry that fails mapping, clock, coverage or health checks; state excluded intervals and residual uncertainty. Filter hard constraints first, compare only eligible options under explicit criteria, show alternatives and never apply changes automatically.

ANSWER BOUNDARY

Do not change capacity, infer attendance, grant rooms or alter pedagogical group size. Programme and room owners approve any restructure. The service cannot create, approve, publish or change the timetable.

WHY THIS IS USEFUL

Shows where a late size change requires immediate action rather than a blanket timetable rebuild.

TT-054

LIFT OUTAGE IMPACT

L3 - recommendation

R2 - integration-dependent

TIMETABLING TEAM QUESTION

If a lift outage removes a verified step-free route, which sessions need a different room, time or authorised contingency?

SENSORS AND TELEMETRY

Use owner-issued lift, door and route-segment status for the complete named route. Proposed anonymous directional counts may supplement only the covered corridor, stair and lift-lobby boundaries, at 1-minute sampling aggregated over 5 minutes. Require commissioned mapping, synchronized clocks, eligible coverage, uptime/gap, drift, maintenance and relocation status; no identities, cameras, device tracking or linked movement histories.

AUTHORITATIVE SUPPORTING DATA

Owner-issued lift outage and restoration estimate; current whole-route accessibility records; affected sessions and destinations; authorised functional requirements; eligible alternative rooms; and official closure or safety notices. Source owners: central timetabling and room owners; programme/HR owner where relevant; IT/AV service; Estates/Facilities; Security and official communications owners. Record source version, effective/observed and retrieved times, permission scope, staleness and conflicts. Current owner-issued records prevail over history, inference or telemetry; equal-authority conflicts remain explicit.

ANALYSIS REQUIRED

Operation: accessibility impact calculation and constrained recommendation. Decision window: outage window. Revalidate every affected route and compare room or time alternatives. If restoration or route status is uncertain, provide time-bounded options and a recheck point. Exclude telemetry that fails mapping, clock, coverage or health checks; state excluded intervals and residual uncertainty. Filter hard constraints first, compare only eligible options under explicit criteria, show alternatives and never apply changes automatically.

ANSWER BOUNDARY

Do not infer disability, disclose adjustment reasons, operate the lift, certify accessibility or issue emergency instructions. Accessibility, facilities and timetable owners retain authority. Do not substitute an uncovered route segment or adjacent boundary for complete route evidence. If mapping, timing, coverage or sensor health is inadequate, exclude the stream; do not impute or substitute nearby data. The service cannot create, approve, publish or change the timetable.

WHY THIS IS USEFUL

Allows accessible teaching arrangements to be reviewed before an outage strands a session.

6. Late changes, disruption and communication

TT-055 to TT-056 | 10 questions in this section | Propagate owner-confirmed changes, compare authorised alternatives and expose cross-system conflicts without operating services.

TT-055

AV OR NETWORK OUTAGE

L3 - recommendation

R2 - integration-dependent

TIMETABLING TEAM QUESTION

If a required AV, capture or network service is unavailable, which authorised room or delivery alternatives remain feasible?

SENSORS AND TELEMETRY

Use only owner-issued non-content status for each named room or served asset: door/lift state, workstation or AV inventory/health, network/power availability, setup state or maintenance flag as relevant. Preserve asset-to-room mapping, event and retrieval times, heartbeat, maintenance mode, permission and conflicts. Do not inspect screens, files, traffic or teaching content.

AUTHORITATIVE SUPPORTING DATA

Current owner-issued service outage and restoration estimate; session's mandatory technology; approved delivery modes; eligible rooms; network and power status; access; timetable constraints; and support ownership. Source owners: central timetabling and room owners; programme/HR owner where relevant; IT/AV service; Estates/Facilities; Security and official communications owners. Record source version, effective/observed and retrieved times, permission scope, staleness and conflicts. Current owner-issued records prevail over history, inference or telemetry; equal-authority conflicts remain explicit.

ANALYSIS REQUIRED

Operation: service-impact lookup, hard-constraint filtering and recommendation. Decision window: outage window. Exclude options lacking a mandatory service, compare equivalent rooms or approved modes and show uncertainty. If restoration status is stale, defer to AV or IT. Exclude telemetry that fails mapping, clock, coverage or health checks; state excluded intervals and residual uncertainty. Filter hard constraints first, compare only eligible options under explicit criteria, show alternatives and never apply changes automatically.

ANSWER BOUNDARY

Do not inspect content, operate equipment, promise service, authorise remote delivery or change the timetable. Academic and service owners approve any alternative. The service cannot create, approve, publish or change the timetable.

WHY THIS IS USEFUL

Identifies realistic alternatives without treating an empty room as technologically suitable.

TT-056

VERIFIED ROOM ENVIRONMENT ISSUE

L3 - diagnosis

R2 - integration-dependent

TIMETABLING TEAM QUESTION

When a room owner confirms an environmental fault, which timetable alternative preserves essential requirements with the least disruption?

SENSORS AND TELEMETRY

Use owner-confirmed room or HVAC fault status plus proposed room-level temperature, relative humidity and direct CO2 sensing in the occupied zone; optional numerical LAeq only where relevant. Use 1-minute observations and 15-minute aggregates with calibration, clock alignment, uptime, gaps and drift. Corridor or adjacent-room readings cannot establish the room condition.

AUTHORITATIVE SUPPORTING DATA

Owner-issued fault or closure status; affected interval and room; current eligible alternatives; capacity, access and equipment requirements; maintenance review time; adjacent sessions; and official safety communication if any. Source owners: central timetabling and room owners; programme/HR owner where relevant; IT/AV service; Estates/Facilities; Security and official communications owners. Record source version, effective/observed and retrieved times, permission scope, staleness and conflicts. Current owner-issued records prevail over history, inference or telemetry; equal-authority conflicts remain explicit.

ANALYSIS REQUIRED

Operation: authorised-status lookup, impact calculation and recommendation. Decision window: confirmed fault window. Treat telemetry as supporting context only, remove any owner-closed room and compare eligible alternatives. If status or coverage is unclear, defer rather than diagnose the fault. Exclude telemetry that fails mapping, clock, coverage or health checks; state excluded intervals and residual uncertainty. Identify the minimum conflict set or temporal association; do not claim causation.

ANSWER BOUNDARY

Do not certify health or safety, diagnose building systems, reopen the room, override maintenance or operate HVAC. The service supports timetable alternatives only. If mapping, timing, coverage or sensor health is inadequate, exclude the stream; do not impute or substitute nearby data. It may report an association, not an unverified cause. The service cannot create, approve, publish or change the timetable.

WHY THIS IS USEFUL

Helps preserve teaching while keeping technical and safety diagnosis with competent owners.

6. Late changes, disruption and communication

TT-057 to TT-058 | 10 questions in this section | Propagate owner-confirmed changes, compare authorised alternatives and expose cross-system conflicts without operating services.

TT-057

SEVERE WEATHER OR CAMPUS DISRUPTION

L3 - comparison

R2 - integration-dependent

TIMETABLING TEAM QUESTION

During an official severe-weather or campus-disruption period, which sessions have authorised remote, relocation or rescheduling options?

SENSORS AND TELEMETRY

No indoor sensor can determine whether teaching should proceed. Use official weather, transport, campus and building status feeds. A commissioned local weather station may provide contextual temperature, wind or rainfall observations with observed/retrieved times, calibration and uptime flags, but it cannot declare the campus open, closed or safe.

AUTHORITATIVE SUPPORTING DATA

Official university and campus communications; current weather and transport authority information; approved disruption policy; session delivery permissions; room and network status; timetable; and accessibility implications. Source owners: central timetabling and room owners; programme/HR owner where relevant; IT/AV service; Estates/Facilities; Security and official communications owners. Record source version, effective/observed and retrieved times, permission scope, staleness and conflicts. Current owner-issued records prevail over history, inference or telemetry; equal-authority conflicts remain explicit.

ANALYSIS REQUIRED

Operation: policy-bounded authorised lookup and scenario comparison. Decision window: the official disruption window. Map each session to permitted alternatives and show unresolved dependencies. If no official status or policy exists, defer and return not assessable. Exclude telemetry that fails mapping, clock, coverage or health checks; state excluded intervals and residual uncertainty. Compare only eligible cases under matched programme, room, time and occupancy conditions; disclose residual confounders.

ANSWER BOUNDARY

Do not declare closure, safety, travel feasibility or remote-teaching approval; operate systems; or override official communications. Competent authorities take precedence. The service cannot create, approve, publish or change the timetable.

WHY THIS IS USEFUL

Allows the team to prepare compliant options without inventing a response to an uncertain event.

TT-058

CHANGE COMMUNICATION SCOPE

L2 - calculation

R2 - integration-dependent

TIMETABLING TEAM QUESTION

For this confirmed timetable change, which authorised channels and recipient groups must be updated, and by what deadline?

SENSORS AND TELEMETRY

No physical sensing is required. Use non-content audit metadata from authorised timetable and communication systems: change/version ID, owner, recipient role-group, publication time, delivery or acknowledgement state, retrieval time and service health. Do not inspect message bodies, personal inboxes or individual response behaviour.

AUTHORITATIVE SUPPORTING DATA

Confirmed change record; authoritative recipient-group rules; module and cohort distribution groups; staff role assignments; publication and notification deadlines; channel ownership; accessibility-format requirements; and system status. Source owners: central timetabling and room owners; programme/HR owner where relevant; IT/AV service; Estates/Facilities; Security and official communications owners. Record source version, effective/observed and retrieved times, permission scope, staleness and conflicts. Current owner-issued records prevail over history, inference or telemetry; equal-authority conflicts remain explicit.

ANALYSIS REQUIRED

Operation: authorised lookup plus recipient-scope and deadline calculation. Decision window: change publication. Produce channel and role groups, not personal content, and verify delivery-system readiness. If the change is not approved, do not prepare a final notice. Validate version, effective time, permission and conflicts; never infer hard constraints from history or telemetry. If minimum evidence is missing, stale, conflicting or unauthorised, qualify the answer or return not assessable.

ANSWER BOUNDARY

Do not send messages, expose personal data, infer who has read them or alter communication policy. The responsible owner publishes the change. The service cannot create, approve, publish or change the timetable.

WHY THIS IS USEFUL

Prevents a confirmed change being published in one system while affected groups remain uninformed.

6. Late changes, disruption and communication

TT-059 to TT-060 | 10 questions in this section | Propagate owner-confirmed changes, compare authorised alternatives and expose cross-system conflicts without operating services.

TT-059

CROSS-SYSTEM CHANGE CONSISTENCY

L3 - diagnosis

R2 - integration-dependent

TIMETABLING TEAM QUESTION

Has the confirmed change propagated consistently across the timetable, room-booking and authorised user-facing systems?

SENSORS AND TELEMETRY

No environmental sensing is required. Use owner-issued record versions and non-content update events from the timetable, room-booking and authorised user-facing systems, preserving source owner, version, effective and retrieved times, propagation status, service health and conflict state. Do not inspect user content.

AUTHORITATIVE SUPPORTING DATA

Approved change identifier and effective time; authoritative timetable; room booking; authorised display or notification systems; system ownership; version identifiers; publication status; and known delays. Source owners: central timetabling and room owners; programme/HR owner where relevant; IT/AV service; Estates/Facilities; Security and official communications owners. Record source version, effective/observed and retrieved times, permission scope, staleness and conflicts. Current owner-issued records prevail over history, inference or telemetry; equal-authority conflicts remain explicit.

ANALYSIS REQUIRED

Operation: cross-system consistency diagnosis. Decision window: the propagation window. Compare key fields and versions, distinguish expected delay from contradiction and identify the owning system. If the authoritative record is unclear, stop publication and defer. Validate version, effective time, permission and conflicts; never infer hard constraints from history or telemetry. Identify the minimum conflict set or temporal association; do not claim causation.

ANSWER BOUNDARY

Do not edit systems, send notices or decide which source wins without the documented authority order. Do not expose personal schedules beyond authorised scope. It may report an association, not an unverified cause. The service cannot create, approve, publish or change the timetable.

WHY THIS IS USEFUL

Finds stale or contradictory timetable information before it causes avoidable confusion.

TT-060

CONFLICTING AUTHORITATIVE SOURCES

L3 - diagnosis

R2 - integration-dependent

TIMETABLING TEAM QUESTION

What limited status can we show when current timetable, room, access or closure owners disagree?

SENSORS AND TELEMETRY

No sensor can resolve a disagreement between authoritative owners. Owner-issued version, effective time, retrieval time and system-health metadata may expose staleness or propagation delay, but telemetry cannot select the correct timetable, room, access or closure fact.

AUTHORITATIVE SUPPORTING DATA

Documented source hierarchy and owners; conflicting records and versions; effective and retrieved times; permission scope; current incident or maintenance notices; and escalation route. Source owners: central timetabling and room owners; programme/HR owner where relevant; IT/AV service; Estates/Facilities; Security and official communications owners. Record source version, effective/observed and retrieved times, permission scope, staleness and conflicts. Current owner-issued records prevail over history, inference or telemetry; equal-authority conflicts remain explicit.

ANALYSIS REQUIRED

Operation: provenance comparison and safe-publication diagnosis. Decision window: until the conflict is resolved. Identify the conflict, apply only documented precedence, suppress unsupported detail and show a recheck or owner contact. Without precedence, return not assessable. Validate version, effective time, permission and conflicts; never infer hard constraints from history or telemetry. Identify the minimum conflict set or temporal association; do not claim causation.

ANSWER BOUNDARY

Do not choose a convenient source, create a timetable fact, grant access or imply safety. The responsible source owner resolves the conflict. It may report an association, not an unverified cause. The service cannot create, approve, publish or change the timetable.

WHY THIS IS USEFUL

Supports honest communication instead of presenting contradictory records as certainty.

7. Fairness, workload, priority and exception handling

TT-061 to TT-062 | 10 questions in this section | Apply explicit approved criteria to aggregate patterns while protecting restricted reasons and never ranking people.

TT-061

EARLY AND LATE SLOT DISTRIBUTION

L3 - comparison

R3 - research or restricted

TIMETABLING TEAM QUESTION

Are early and late teaching slots distributed consistently across comparable cohorts under the approved fairness criteria?

SENSORS AND TELEMETRY

No physical sensing is needed for the core result. Where building-open, closure or service status matters, use only the current owner-issued feed for the named asset, with verified mapping, observed/retrieved time, heartbeat, maintenance state and conflicts. Telemetry does not create permission or availability.

AUTHORITATIVE SUPPORTING DATA

Current draft timetable; explicit authorised definition of early and late; comparable cohort grouping; programme delivery constraints; part-time patterns; approved exceptions; and governance permission for aggregate fairness review. Source owners: academic-policy, employment/workload and accessibility owners; programme owner; central timetabling and room owners. Record source version, effective/observed and retrieved times, permission scope, staleness and conflicts. Current owner-issued records prevail over history, inference or telemetry; equal-authority conflicts remain explicit.

ANALYSIS REQUIRED

Operation: aggregate fairness comparison. Decision window: term or year. Compare slot shares within eligible peer groups, show exposure counts and sensitivity to the grouping rule, and separate authorised exceptions. If criteria or comparators are disputed, return not assessable. Validate version, effective time, permission and conflicts; never infer hard constraints from history or telemetry. Compare only eligible cases under matched programme, room, time and occupancy conditions; disclose residual confounders.

ANSWER BOUNDARY

Do not infer hardship or protected traits, rank cohorts or decide what is fair from statistics alone. Programme and governance owners interpret the review. The service cannot create, approve, publish or change the timetable.

WHY THIS IS USEFUL

Provides transparent evidence for checking whether undesirable time bands are concentrated in one pattern.

TT-062

STAFF TEACHING SPAN

L3 - calculation

R3 - research or restricted

TIMETABLING TEAM QUESTION

Which teaching days exceed authorised span, consecutive-session or break constraints for assigned staff?

SENSORS AND TELEMETRY

No physical sensing is needed for the core result. Where building-open, closure or service status matters, use only the current owner-issued feed for the named asset, with verified mapping, observed/retrieved time, heartbeat, maintenance state and conflicts. Telemetry does not create permission or availability.

AUTHORITATIVE SUPPORTING DATA

Permission-bounded staff teaching schedule; authorised span, break and consecutive-session rules; approved availability constraints; room locations and transition allowances; and assignment ownership. Source owners: academic-policy, employment/workload and accessibility owners; programme owner; central timetabling and room owners. Record source version, effective/observed and retrieved times, permission scope, staleness and conflicts. Current owner-issued records prevail over history, inference or telemetry; equal-authority conflicts remain explicit.

ANALYSIS REQUIRED

Operation: workload-constraint calculation. Decision window: each teaching day. Calculate first-to-last span, consecutive teaching, protected breaks and transitions, and return only rule breaches needed for timetable action. Missing permission or rule ownership requires deferral. Validate version, effective time, permission and conflicts; never infer hard constraints from history or telemetry. If minimum evidence is missing, stale, conflicting or unauthorised, qualify the answer or return not assessable.

ANSWER BOUNDARY

Do not infer total workload, productivity, wellbeing or reasons for a constraint, rank staff, or disclose personal schedules beyond authorised need. Employment and academic owners decide remedies. The service cannot create, approve, publish or change the timetable.

WHY THIS IS USEFUL

Finds timetable-specific breaches without turning the service into a staff-performance system.

7. Fairness, workload, priority and exception handling

TT-063 to TT-064 | 10 questions in this section | Apply explicit approved criteria to aggregate patterns while protecting restricted reasons and never ranking people.

TT-063

AUTHORISED ADJUSTMENT CONSTRAINTS

L3 - calculation

R3 - research or restricted

TIMETABLING TEAM QUESTION Does the draft respect every authorised scheduling constraint for reasonable adjustments, contractual needs or agreed working patterns?

SENSORS AND TELEMETRY

No physical sensing is needed for the core result. Where building-open, closure or service status matters, use only the current owner-issued feed for the named asset, with verified mapping, observed/retrieved time, heartbeat, maintenance state and conflicts. Telemetry does not create permission or availability.

AUTHORITATIVE SUPPORTING DATA

Permission-bounded constraint tokens or usable windows from the responsible employment or accessibility owner; assigned sessions; draft timetable; route or room requirements; and exception approval. Reasons or diagnoses are excluded. Source owners: academic-policy, employment/workload and accessibility owners; programme owner; central timetabling and room owners. Record source version, effective/observed and retrieved times, permission scope, staleness and conflicts. Current owner-issued records prevail over history, inference or telemetry; equal-authority conflicts remain explicit.

ANALYSIS REQUIRED

Operation: compliance-style constraint check, not legal certification. Decision window: draft review. Test only the authorised scheduling representation, report pass, conflict or missing data and identify affected sessions. If the requester lacks permission, return restricted. Validate version, effective time, permission and conflicts; never infer hard constraints from history or telemetry. If minimum evidence is missing, stale, conflicting or unauthorised, qualify the answer or return not assessable.

ANSWER BOUNDARY

Do not request or reveal diagnoses or reasons, determine eligibility, reinterpret an adjustment, or certify legal compliance. Responsible owners retain authority. The service cannot create, approve, publish or change the timetable.

WHY THIS IS USEFUL

Allows the team to detect scheduling conflicts while minimising access to sensitive information.

TT-064

PART-TIME PROGRAMME PATTERN

L3 - comparison

R2 - integration-dependent

TIMETABLING TEAM QUESTION Are part-time cohorts spread across more days or longer gaps than the authorised programme pattern requires?

SENSORS AND TELEMETRY

No physical sensing is needed for the core result. Where building-open, closure or service status matters, use only the current owner-issued feed for the named asset, with verified mapping, observed/retrieved time, heartbeat, maintenance state and conflicts. Telemetry does not create permission or availability.

AUTHORITATIVE SUPPORTING DATA

Current part-time programme pattern; authorised cohort grouping; draft timetable; compulsory activities; room and staff constraints; route allowances; and approved exceptions. Source owners: academic-policy, employment/workload and accessibility owners; programme owner; central timetabling and room owners. Record source version, effective/observed and retrieved times, permission scope, staleness and conflicts. Current owner-issued records prevail over history, inference or telemetry; equal-authority conflicts remain explicit.

ANALYSIS REQUIRED

Operation: pattern comparison. Decision window: each teaching week. Calculate attendance days, day spans and unusable gaps, compare with the authorised pattern and show which hard constraints caused any deviation. If the pattern is advisory only, label it as a preference. Validate version, effective time, permission and conflicts; never infer hard constraints from history or telemetry. Compare only eligible cases under matched programme, room, time and occupancy conditions; disclose residual confounders.

ANSWER BOUNDARY

Do not infer employment, caring or personal circumstances, rank cohorts or relax academic requirements. Programme owners decide acceptable exceptions. The service cannot create, approve, publish or change the timetable.

WHY THIS IS USEFUL

Supports a timetable that respects the published structure of part-time study.

7. Fairness, workload, priority and exception handling

TT-065 to TT-066 | 10 questions in this section | Apply explicit approved criteria to aggregate patterns while protecting restricted reasons and never ranking people.

TT-065

SCARCE SPECIALIST ROOM PRIORITY

L3 - recommendation

R3 - research or restricted

TIMETABLING TEAM QUESTION

When one specialist room is scarce, which sessions have an authorised need that cannot be met elsewhere?

SENSORS AND TELEMETRY

Use only owner-issued non-content status for each named room or served asset: door/lift state, workstation or AV inventory/health, network/power availability, setup state or maintenance flag as relevant. Preserve asset-to-room mapping, event and retrieval times, heartbeat, maintenance mode, permission and conflicts. Do not inspect screens, files, traffic or teaching content.

AUTHORITATIVE SUPPORTING DATA

Current specialist-room eligibility and status; each activity's essential equipment, safety and accessibility requirements; equivalent-room inventory; approved priority criteria; cohort and staff windows; and owner decisions. Source owners: academic-policy, employment/workload and accessibility owners; programme owner; central timetabling and room owners. Record source version, effective/observed and retrieved times, permission scope, staleness and conflicts. Current owner-issued records prevail over history, inference or telemetry; equal-authority conflicts remain explicit.

ANALYSIS REQUIRED

Operation: hard-constraint classification and decision-support recommendation. Decision window: allocation conflict. Identify sessions with no eligible substitute, apply only explicit priority criteria and show unresolved ties. If criteria are absent, present the conflict without ranking. Exclude telemetry that fails mapping, clock, coverage or health checks; state excluded intervals and residual uncertainty. Filter hard constraints first, compare only eligible options under explicit criteria, show alternatives and never apply changes automatically.

ANSWER BOUNDARY

Do not invent priority, rank staff or programmes, grant the room, relax safety or accessibility requirements, or infer importance from enrolment alone. Authorised owners decide. The service cannot create, approve, publish or change the timetable.

WHY THIS IS USEFUL

Makes a scarce-resource conflict transparent and focuses attention on sessions with no genuine alternative.

TT-066

DISTRIBUTION OF TRAVEL BURDEN

L3 - comparison

R3 - research or restricted

TIMETABLING TEAM QUESTION

Are avoidable campus transitions distributed unevenly across comparable cohorts or teaching teams?

SENSORS AND TELEMETRY

No physical sensing is needed for the core result. Proposed anonymous directional counters may qualify recurrent congestion only at the exact corridor, stair or lift-lobby boundaries used by the route, using 1-minute observations and 5-minute aggregates. Require commissioned coverage, synchronized clocks, uptime/gap, drift and relocation status; no identities or path histories.

AUTHORITATIVE SUPPORTING DATA

Current draft timetable; verified room and campus locations; route and transition model; authorised comparator groups; accessibility constraints; programme patterns; staff assignment permissions; and approved exceptions. Source owners: academic-policy, employment/workload and accessibility owners; programme owner; central timetabling and room owners. Record source version, effective/observed and retrieved times, permission scope, staleness and conflicts. Current owner-issued records prevail over history, inference or telemetry; equal-authority conflicts remain explicit.

ANALYSIS REQUIRED

Operation: aggregate distribution comparison. Decision window: term or year. Calculate scheduled transition count and time for eligible peer groups, test sensitivity to route assumptions and separate required from avoidable travel. If comparators are not agreed, return not assessable. Exclude telemetry that fails mapping, clock, coverage or health checks; state excluded intervals and residual uncertainty. Compare only eligible cases under matched programme, room, time and occupancy conditions; disclose residual confounders.

ANSWER BOUNDARY

Do not track individuals, infer hardship, rank people or claim inequity from travel distance alone. Governance and programme owners interpret the result. Do not infer individual attendance, absence, engagement, performance or reasons from aggregate use. Do not substitute an uncovered route segment or adjacent boundary for complete route evidence. The service cannot create, approve, publish or change the timetable.

WHY THIS IS USEFUL

Provides evidence for redistributing unnecessary travel without compromising essential teaching.

7. Fairness, workload, priority and exception handling

TT-067 to TT-068 | 10 questions in this section | Apply explicit approved criteria to aggregate patterns while protecting restricted reasons and never ranking people.

TT-067

CONSISTENT EXCEPTION HANDLING

L3 - comparison

R2 - integration-dependent

TIMETABLING TEAM QUESTION

Are similar timetable exceptions handled consistently under the same authorised rule?

SENSORS AND TELEMETRY

No physical sensing is needed for the core result. Where building-open, closure or service status matters, use only the current owner-issued feed for the named asset, with verified mapping, observed/retrieved time, heartbeat, maintenance state and conflicts. Telemetry does not create permission or availability.

AUTHORITATIVE SUPPORTING DATA

Permission-controlled exception records with reasons reduced to rule categories; decision owner; effective period; module or programme context; approved precedent; and current timetable outcome. Source owners: academic-policy, employment/workload and accessibility owners; programme owner; central timetabling and room owners. Record source version, effective/observed and retrieved times, permission scope, staleness and conflicts. Current owner-issued records prevail over history, inference or telemetry; equal-authority conflicts remain explicit.

ANALYSIS REQUIRED

Operation: rule-based audit comparison. Decision window: timetable cycle. Group only genuinely comparable cases, compare applied rule and outcome, and flag unexplained differences for owner review. Small or sensitive groups are suppressed. Validate version, effective time, permission and conflicts; never infer hard constraints from history or telemetry. Compare only eligible cases under matched programme, room, time and occupancy conditions; disclose residual confounders.

ANSWER BOUNDARY

Do not expose personal circumstances, infer protected traits, overturn decisions or create precedent. Responsible owners explain or amend an exception. The service cannot create, approve, publish or change the timetable.

WHY THIS IS USEFUL

Helps the team apply documented rules consistently without disclosing why an individual exception exists.

TT-068

HIGH-INTENSITY SEQUENCES

L3 - comparison

R2 - integration-dependent

TIMETABLING TEAM QUESTION

Are practicals, presentations or assessments placed back-to-back without the authorised preparation, reset or recovery interval?

SENSORS AND TELEMETRY

No physical sensing is needed for the core result. Where building-open, closure or service status matters, use only the current owner-issued feed for the named asset, with verified mapping, observed/retrieved time, heartbeat, maintenance state and conflicts. Telemetry does not create permission or availability.

AUTHORITATIVE SUPPORTING DATA

Approved activity-intensity category and required interval; draft timetable; room transition and reset time; cohort and staff sequences; accessibility requirements; and authorised exceptions. Source owners: academic-policy, employment/workload and accessibility owners; programme owner; central timetabling and room owners. Record source version, effective/observed and retrieved times, permission scope, staleness and conflicts. Current owner-issued records prevail over history, inference or telemetry; equal-authority conflicts remain explicit.

ANALYSIS REQUIRED

Operation: sequence-risk calculation and comparison. Decision window: each affected day. Identify consecutive high-intensity activities, subtract travel and setup time and compare with the approved interval. If the interval is not defined, ask the academic owner rather than invent one. Validate version, effective time, permission and conflicts; never infer hard constraints from history or telemetry. Compare only eligible cases under matched programme, room, time and occupancy conditions; disclose residual confounders.

ANSWER BOUNDARY

Do not infer stress, health, learning or performance, or decide that an activity is intensive from content. Use the approved category only. The service cannot create, approve, publish or change the timetable.

WHY THIS IS USEFUL

Finds demanding sequences that need spacing even when no direct time clash exists.

7. Fairness, workload, priority and exception handling

TT-069 to TT-070 | 10 questions in this section | Apply explicit approved criteria to aggregate patterns while protecting restricted reasons and never ranking people.

TT-069

HARD CONSTRAINTS VS PREFERENCES

L3 - recommendation

R2 - integration-dependent

TIMETABLING TEAM QUESTION

How should a conflict between hard constraints and preferences be presented so the authorised owner can resolve it transparently?

SENSORS AND TELEMETRY

No physical sensing is needed for the core result. Where building-open, closure or service status matters, use only the current owner-issued feed for the named asset, with verified mapping, observed/retrieved time, heartbeat, maintenance state and conflicts. Telemetry does not create permission or availability.

AUTHORITATIVE SUPPORTING DATA

Current constraint registry; owner and precedence; hard or preference classification; affected sessions; available options; permission; and any approved exception process. Source owners: academic-policy, employment/workload and accessibility owners; programme owner; central timetabling and room owners. Record source version, effective/observed and retrieved times, permission scope, staleness and conflicts. Current owner-issued records prevail over history, inference or telemetry; equal-authority conflicts remain explicit.

ANALYSIS REQUIRED

Operation: constraint explanation and option comparison. Decision window: conflict resolution. Show which options satisfy all hard constraints, which preferences each option misses and the effect of any authorised relaxation. If classification is unclear, stop and seek owner clarification. Validate version, effective time, permission and conflicts; never infer hard constraints from history or telemetry. Filter hard constraints first, compare only eligible options under explicit criteria, show alternatives and never apply changes automatically.

ANSWER BOUNDARY

Do not downgrade a hard constraint, promote a preference, infer reasons, rank people or choose the final option. The service explains choices; the authorised owner decides. The service cannot create, approve, publish or change the timetable.

WHY THIS IS USEFUL

Supports an auditable decision instead of hiding trade-offs inside an optimisation score.

TT-070

FAIRNESS ANSWERABILITY

L3 - answerability

R3 - research or restricted

TIMETABLING TEAM QUESTION

Which proposed fairness checks are responsibly answerable with current authorised data and permissions, and which are not?

SENSORS AND TELEMETRY

No physical sensing is needed for the core result. Where building-open, closure or service status matters, use only the current owner-issued feed for the named asset, with verified mapping, observed/retrieved time, heartbeat, maintenance state and conflicts. Telemetry does not create permission or availability.

AUTHORITATIVE SUPPORTING DATA

Explicit fairness question and criterion; authorised aggregate schedule data; comparator definition; relevant constraints and exceptions; permission and suppression thresholds; provenance; and governance owner. Source owners: academic-policy, employment/workload and accessibility owners; programme owner; central timetabling and room owners. Record source version, effective/observed and retrieved times, permission scope, staleness and conflicts. Current owner-issued records prevail over history, inference or telemetry; equal-authority conflicts remain explicit.

ANALYSIS REQUIRED

Operation: evidence and permission sufficiency assessment. Decision window: proposed audit. Check whether the outcome, comparator, coverage and lawful permission are adequate; classify answerable, qualified, restricted or not assessable, with missing evidence listed. Validate version, effective time, permission and conflicts; never infer hard constraints from history or telemetry. Classify the result as assessable, qualified or not assessable against predeclared thresholds.

ANSWER BOUNDARY

Do not infer protected traits, use proxies for sensitive characteristics, expose small groups, rank people or claim fairness from one metric. Governance owners define and interpret criteria. The service cannot create, approve, publish or change the timetable.

WHY THIS IS USEFUL

Prevents the system from producing confident fairness claims that the available evidence cannot support.

8. Scenario planning, utilisation and lower-impact scheduling

TT-071 to TT-072 | 10 questions in this section | Test transparent, reversible scenarios against hard constraints, uncertainty, service capacity and lower-impact objectives.

TT-071

ENROLMENT GROWTH SCENARIO

L3 - forecast

R2 - integration-dependent

TIMETABLING TEAM QUESTION

Under approved enrolment-growth scenarios, which modules, group structures and room allocations become infeasible first?

SENSORS AND TELEMETRY

No physical sensing is needed for the core calculation. Proposed anonymous room-level doorway or ceiling counts may be used only as secondary historical evidence, with complete entrance coverage, commissioned error bounds, aligned clocks, 1-minute observations aggregated to 15 minutes, and uptime, gap, drift and relocation flags. Counts never replace enrolment, capacity or booking records.

AUTHORITATIVE SUPPORTING DATA

Approved enrolment scenarios and horizon; current module pattern; group-size rules; room capacities and eligibility; staff and specialist-resource constraints; accessibility; and planned room changes. Source owners: planning/enrolment and programme owners; central timetabling and room owners; Estates/Energy; accessibility and route owners; sensor-data steward. Record source version, effective/observed and retrieved times, permission scope, staleness and conflicts. Current owner-issued records prevail over history, inference or telemetry; equal-authority conflicts remain explicit.

ANALYSIS REQUIRED

Operation: scenario forecast and threshold analysis. Decision window: future timetable cycle. Recalculate group and room demand across the authorised range, identify first constraint breaches and show sensitivity to room or staff assumptions. Report ranges, not false precision. Exclude telemetry that fails mapping, clock, coverage or health checks; state excluded intervals and residual uncertainty. Use eligible comparators, report ranges and sensitivity, and return not assessable when coverage or drift is inadequate.

ANSWER BOUNDARY

Do not predict named applicants or enrolment, change capacity, assign resources or publish a timetable. Planning and academic owners approve assumptions and action. Do not infer individual attendance, absence, engagement, performance or reasons from aggregate use. The service cannot create, approve, publish or change the timetable.

WHY THIS IS USEFUL

Shows where future growth needs structural timetable or space planning rather than late reactive changes.

TT-072

LOSS OF A LARGE ROOM

L3 - simulation

R2 - integration-dependent

TIMETABLING TEAM QUESTION

What happens to the draft timetable if one large teaching room is unavailable for the stated period?

SENSORS AND TELEMETRY

Use only owner-issued non-content status for each named room or served asset: door/lift state, workstation or AV inventory/health, network/power availability, setup state or maintenance flag as relevant. Preserve asset-to-room mapping, event and retrieval times, heartbeat, maintenance mode, permission and conflicts. Do not inspect screens, files, traffic or teaching content.

AUTHORITATIVE SUPPORTING DATA

Scenario room and outage interval; current draft; eligible alternative rooms; capacity, layout, equipment and accessibility requirements; adjacent sessions; specialist resources; and publication constraints. Source owners: planning/enrolment and programme owners; central timetabling and room owners; Estates/Energy; accessibility and route owners; sensor-data steward. Record source version, effective/observed and retrieved times, permission scope, staleness and conflicts. Current owner-issued records prevail over history, inference or telemetry; equal-authority conflicts remain explicit.

ANALYSIS REQUIRED

Operation: resilience simulation. Decision window: specified outage. Remove the room, re-solve feasible placements without relaxing hard constraints and report displaced sessions, bottlenecks and unresolved cases. If the outage period or room inventory is uncertain, run bounded scenarios. Exclude telemetry that fails mapping, clock, coverage or health checks; state excluded intervals and residual uncertainty. Use eligible comparators, report ranges and sensitivity, and return not assessable when coverage or drift is inadequate.

ANSWER BOUNDARY

Do not declare a real closure, grant alternatives, relax capacity or safety, or publish a revised timetable. The result is a planning scenario for authorised owners. The service cannot create, approve, publish or change the timetable.

WHY THIS IS USEFUL

Reveals dependence on a scarce large room and supports contingency planning.

8. Scenario planning, utilisation and lower-impact scheduling

TT-073 to TT-074 | 10 questions in this section | Test transparent, reversible scenarios against hard constraints, uncertainty, service capacity and lower-impact objectives.

TT-073

ROOM RIGHT-SIZING AND ENERGY

L3 - recommendation

R3 - research or restricted

TIMETABLING TEAM QUESTION

Could eligible sessions move to better-sized rooms to reduce energy demand without weakening access, equipment, environmental quality or transitions?

SENSORS AND TELEMETRY

Proposed calibrated electrical submeters and BMS demand points mapped to the room, lighting circuit and served HVAC zone, with 1-minute demand and 15-minute energy aggregates; optional anonymous room counts and room-level environment streams. Require verified meter multipliers, commissioning, synchronized clocks, uptime/gap, drift, maintenance and asset provenance.

AUTHORITATIVE SUPPORTING DATA

Current eligible-room inventory; capacity and layout; accessibility and equipment; timetable; room or zone energy mapping; BMS operation; comparable anonymous use; weather or season; and explicit carbon or energy objective. Source owners: planning/enrolment and programme owners; central timetabling and room owners; Estates/Energy; accessibility and route owners; sensor-data steward. Record source version, effective/observed and retrieved times, permission scope, staleness and conflicts. Current owner-issued records prevail over history, inference or telemetry; equal-authority conflicts remain explicit.

ANALYSIS REQUIRED

Operation: matched energy comparison and multi-objective recommendation. Decision window: future timetable scenario. Filter for full eligibility first, estimate energy difference under comparable conditions and report trade-offs and uncertainty. If submeter or served-zone mapping is absent, do not claim savings. Exclude telemetry that fails mapping, clock, coverage or health checks; state excluded intervals and residual uncertainty. Filter hard constraints first, compare only eligible options under explicit criteria, show alternatives and never apply changes automatically.

ANSWER BOUNDARY

Do not operate HVAC or lighting, certify comfort or carbon savings, infer attendance or trade away hard constraints. Energy evidence is advisory and owners approve moves. If mapping, timing, coverage or sensor health is inadequate, exclude the stream; do not impute or substitute nearby data. The service cannot create, approve, publish or change the timetable.

WHY THIS IS USEFUL

Identifies lower-demand room choices that remain educationally and operationally suitable.

TT-074

CONSOLIDATED TEACHING DAYS

L3 - comparison

R3 - research or restricted

TIMETABLING TEAM QUESTION

Would consolidating eligible teaching onto fewer days reduce travel or building demand, and what new constraints or inequities would it create?

SENSORS AND TELEMETRY

Proposed aggregate building or zone energy and demand telemetry may qualify scenarios only where room/HVAC mapping, meter commissioning, 15-minute aggregation, clock alignment, uptime, gaps, drift and maintenance state are verified. Use anonymous room-use evidence only at aggregate level; do not track individual travel.

AUTHORITATIVE SUPPORTING DATA

Current programme patterns; staff and cohort constraints; room eligibility; route model; accessibility; protected breaks; energy or building-demand model; approved comparator groups; and scenario assumptions. Source owners: planning/enrolment and programme owners; central timetabling and room owners; Estates/Energy; accessibility and route owners; sensor-data steward. Record source version, effective/observed and retrieved times, permission scope, staleness and conflicts. Current owner-issued records prevail over history, inference or telemetry; equal-authority conflicts remain explicit.

ANALYSIS REQUIRED

Operation: constrained scenario comparison. Decision window: future timetable cycle. Compare days attended, span, travel, room demand and energy proxy while preserving every hard constraint; show distributional effects and sensitivity. If fairness criteria are absent, do not declare a preferred scenario. Exclude telemetry that fails mapping, clock, coverage or health checks; state excluded intervals and residual uncertainty. Compare only eligible cases under matched programme, room, time and occupancy conditions; disclose residual confounders.

ANSWER BOUNDARY

Do not infer personal circumstances, rank people, change programme delivery or claim guaranteed energy savings. Authorised academic and governance owners decide. Do not infer individual attendance, absence, engagement, performance or reasons from aggregate use. If mapping, timing, coverage or sensor health is inadequate, exclude the stream; do not impute or substitute nearby data. The service cannot create, approve, publish or change the timetable.

WHY THIS IS USEFUL

Makes the trade-offs of concentrated teaching visible before a policy or pattern is proposed.

8. Scenario planning, utilisation and lower-impact scheduling

TT-075 to TT-076 | 10 questions in this section | Test transparent, reversible scenarios against hard constraints, uncertainty, service capacity and lower-impact objectives.

TT-075

STAGGERED START-TIME SCENARIO

L3 - forecast

R3 - research or restricted

TIMETABLING TEAM QUESTION

Which alternative start-time pattern is most likely to reduce class-change congestion while preserving teaching, travel and accessibility requirements?

SENSORS AND TELEMETRY

Proposed anonymous directional counts at every relevant corridor, stair, lift-lobby or doorway boundary; 1-minute observations aggregated over 5 minutes. Require complete boundary coverage, commissioning against reference counts, synchronized clocks, uptime/gap, drift, maintenance and relocation status. No identities, cameras, device tracking or linked movement histories.

AUTHORITATIVE SUPPORTING DATA

Current draft; room capacities and exits; route graph; covered aggregate counts; lift or door status; session-duration rules; protected breaks; transport and staff constraints; and approved scenario limits. Source owners: planning/enrolment and programme owners; central timetabling and room owners; Estates/Energy; accessibility and route owners; sensor-data steward. Record source version, effective/observed and retrieved times, permission scope, staleness and conflicts. Current owner-issued records prevail over history, inference or telemetry; equal-authority conflicts remain explicit.

ANALYSIS REQUIRED

Operation: flow forecast and constrained scenario comparison. Decision window: selected changeover periods. Model timetable demand, calibrate with comparable covered counts and compare stagger patterns with confidence intervals. Uncovered routes or weak calibration require a timetable-only estimate. Exclude telemetry that fails mapping, clock, coverage or health checks; state excluded intervals and residual uncertainty. Use eligible comparators, report ranges and sensitivity, and return not assessable when coverage or drift is inadequate.

ANSWER BOUNDARY

Do not track people, certify safety, operate access systems or change the timetable automatically. Route and timetable owners review the scenario. Incomplete route coverage or failed sensor-quality checks require a qualified or not-assessable result. The service cannot create, approve, publish or change the timetable.

WHY THIS IS USEFUL

Supports targeted staggering instead of moving many sessions without evidence.

TT-076

DAYLIGHT-AWARE SCHEDULING

L3 - comparison

R3 - research or restricted

TIMETABLING TEAM QUESTION

Could more eligible teaching be placed in daylight hours without creating unfair early or late patterns, room conflicts or weaker access?

SENSORS AND TELEMETRY

Proposed room-level illuminance in the occupied/task zone plus lighting-circuit energy mapped to the same room; 1-minute observations and 15-minute aggregates. Require sensor calibration, placement and obstruction notes, daylight/season provenance, clock alignment, uptime, gaps and drift. Aggregate occupancy may contextualise use but cannot identify attendance.

AUTHORITATIVE SUPPORTING DATA

Current draft; eligible room set; academic and staff constraints; authorised fairness criteria; daylight or solar context; lighting control and energy mapping; room access hours; and comparable seasonal periods. Source owners: planning/enrolment and programme owners; central timetabling and room owners; Estates/Energy; accessibility and route owners; sensor-data steward. Record source version, effective/observed and retrieved times, permission scope, staleness and conflicts. Current owner-issued records prevail over history, inference or telemetry; equal-authority conflicts remain explicit.

ANALYSIS REQUIRED

Operation: constrained comparison. Decision window: future timetable scenario. Filter for full eligibility, compare daylight and lighting-demand evidence across comparable seasons and audit early/late distribution. If room-level metering or mapping is absent, report a qualitative scenario only. Exclude telemetry that fails mapping, clock, coverage or health checks; state excluded intervals and residual uncertainty. Compare only eligible cases under matched programme, room, time and occupancy conditions; disclose residual confounders.

ANSWER BOUNDARY

Do not guarantee daylight quality, operate lighting, infer individual preference or trade away access and employment constraints. The scenario is advisory. If mapping, timing, coverage or sensor health is inadequate, exclude the stream; do not impute or substitute nearby data. The service cannot create, approve, publish or change the timetable.

WHY THIS IS USEFUL

Tests a lower-demand scheduling idea while exposing its academic and fairness trade-offs.

8. Scenario planning, utilisation and lower-impact scheduling

TT-077 to TT-078 | 10 questions in this section | Test transparent, reversible scenarios against hard constraints, uncertainty, service capacity and lower-impact objectives.

TT-077

HOURLY TEACHING-SPACE DEMAND

L3 - forecast

R2 - integration-dependent

TIMETABLING TEAM QUESTION

What hourly teaching-space demand is the draft likely to create across Abacws by room type and capacity band?

SENSORS AND TELEMETRY

No physical sensing is needed for the core calculation. Proposed anonymous room-level doorway or ceiling counts may be used only as secondary historical evidence, with complete entrance coverage, commissioned error bounds, aligned clocks, 1-minute observations aggregated to 15 minutes, and uptime, gap, drift and relocation flags. Counts never replace enrolment, capacity or booking records.

AUTHORITATIVE SUPPORTING DATA

Current draft timetable; room type, capacity and eligible hours; setup and closure windows; activity requirements; planned room outages; authorised enrolment range; and any quality-controlled aggregate use history. Source owners: planning/enrolment and programme owners; central timetabling and room owners; Estates/Energy; accessibility and route owners; sensor-data steward. Record source version, effective/observed and retrieved times, permission scope, staleness and conflicts. Current owner-issued records prevail over history, inference or telemetry; equal-authority conflicts remain explicit.

ANALYSIS REQUIRED

Operation: demand calculation plus scenario forecast. Decision window: each timetable week. Sum concurrent booked demand by eligible room class, test enrolment and outage scenarios and report uncertainty. Do not present scheduled demand as observed use. Exclude telemetry that fails mapping, clock, coverage or health checks; state excluded intervals and residual uncertainty. Use eligible comparators, report ranges and sensitivity, and return not assessable when coverage or drift is inadequate.

ANSWER BOUNDARY

Do not infer individual attendance, create room supply, grant bookings or claim a capacity shortage without current room data. Estates and room owners address supply decisions. The service cannot create, approve, publish or change the timetable.

WHY THIS IS USEFUL

Shows where the draft creates peak pressure on particular room types.

TT-078

ROOM UTILISATION SCENARIO

L3 - calculation

R2 - integration-dependent

TIMETABLING TEAM QUESTION

Which room classes are over- or under-used in the draft once eligible hours, capacity, setup and activity requirements are considered?

SENSORS AND TELEMETRY

No physical sensing is needed for the core calculation. Proposed anonymous room-level doorway or ceiling counts may be used only as secondary historical evidence, with complete entrance coverage, commissioned error bounds, aligned clocks, 1-minute observations aggregated to 15 minutes, and uptime, gap, drift and relocation flags. Counts never replace enrolment, capacity or booking records.

AUTHORITATIVE SUPPORTING DATA

Current draft and room bookings; current room inventory; eligible hours; capacity by layout; setup and closure periods; room type; activity requirements; and optional quality-controlled aggregate use. Source owners: planning/enrolment and programme owners; central timetabling and room owners; Estates/Energy; accessibility and route owners; sensor-data steward. Record source version, effective/observed and retrieved times, permission scope, staleness and conflicts. Current owner-issued records prevail over history, inference or telemetry; equal-authority conflicts remain explicit.

ANALYSIS REQUIRED

Operation: scheduled-utilisation calculation and optional realised-use comparison. Decision window: term or year. Use available eligible hours as denominator, separate room classes and report both booked and observed measures with coverage. Missing inventory makes the ratio not assessable. Exclude telemetry that fails mapping, clock, coverage or health checks; state excluded intervals and residual uncertainty. If minimum evidence is missing, stale, conflicting or unauthorised, qualify the answer or return not assessable.

ANSWER BOUNDARY

Do not equate low occupancy with availability, infer attendance, change room purpose or optimise solely for a higher percentage. Room suitability and resilience remain hard constraints. The service cannot create, approve, publish or change the timetable.

WHY THIS IS USEFUL

Provides a defensible utilisation view that does not reward inappropriate overloading.

8. Scenario planning, utilisation and lower-impact scheduling

TT-079 to TT-080 | 10 questions in this section | Test transparent, reversible scenarios against hard constraints, uncertainty, service capacity and lower-impact objectives.

TT-079

ROBUSTNESS TO DATA GAPS

L3 - robustness

R3 - research or restricted

TIMETABLING TEAM QUESTION

How stable is each recommendation when room, route, aggregate-use or service evidence is missing, delayed or wrong?

SENSORS AND TELEMETRY

Use the proposed evidence inventory and quality metadata rather than substituting observations: exact room/route/asset mapping, deployment and relocation history, commissioning/calibration, clock offset, last heartbeat, uptime, gaps, drift, maintenance and permission state. Where aggregate counts or environment streams are relevant, retain their documented error and eligible coverage.

AUTHORITATIVE SUPPORTING DATA

Current recommendation inputs and provenance graph; authoritative-source hierarchy; expected update intervals; permission scope; alternative sources; accepted quality thresholds; and decision owner. Source owners: planning/enrolment and programme owners; central timetabling and room owners; Estates/Energy; accessibility and route owners; sensor-data steward. Record source version, effective/observed and retrieved times, permission scope, staleness and conflicts. Current owner-issued records prevail over history, inference or telemetry; equal-authority conflicts remain explicit.

ANALYSIS REQUIRED

Operation: robustness and sensitivity assessment. Decision window: the recommendation under review. Remove or perturb each uncertain source within defensible bounds, recompute options and report which conclusions survive. If a hard constraint depends on missing data, return not assessable. Exclude telemetry that fails mapping, clock, coverage or health checks; state excluded intervals and residual uncertainty. Use eligible comparators, report ranges and sensitivity, and return not assessable when coverage or drift is inadequate.

ANSWER BOUNDARY

Do not hide uncertainty, impute authority, choose the most convenient source or publish a recommendation that fails under plausible data error. Owners resolve critical gaps. The service cannot create, approve, publish or change the timetable.

WHY THIS IS USEFUL

Shows whether a proposed timetable decision is evidence-resilient or too dependent on fragile data.

TT-080

REVERSIBLE TIMETABLE PILOT

L3 - recommendation

R3 - research or restricted

TIMETABLING TEAM QUESTION

Which limited timetable change is suitable for a reversible pilot, and what evidence would show whether it helped?

SENSORS AND TELEMETRY

Use only the pre-specified proposed aggregate streams needed for the pilot - room counts, route flow, room environment or energy - with exact spatial/asset mapping, baseline and intervention windows, sampling/aggregation, calibration/commissioning, clock alignment, uptime, gaps, drift and privacy controls. No identities, content inspection or undeclared secondary use.

AUTHORITATIVE SUPPORTING DATA

Approved pilot scope and owner; baseline timetable; academic, employment and accessibility constraints; room and service status; success criteria; comparable control periods or groups; feedback protocol; stop conditions; and governance approval. Source owners: planning/enrolment and programme owners; central timetabling and room owners; Estates/Energy; accessibility and route owners; sensor-data steward. Record source version, effective/observed and retrieved times, permission scope, staleness and conflicts. Current owner-issued records prevail over history, inference or telemetry; equal-authority conflicts remain explicit.

ANALYSIS REQUIRED

Operation: pilot-design and evaluation recommendation. Decision window: defined pilot period. Select a low-risk reversible change, pre-register outcomes and confounders, compare with a defensible baseline and classify supported, conflicting or not assessable. State uncertainty and reversion trigger. Exclude telemetry that fails mapping, clock, coverage or health checks; state excluded intervals and residual uncertainty. Filter hard constraints first, compare only eligible options under explicit criteria, show alternatives and never apply changes automatically.

ANSWER BOUNDARY

Do not experiment on access, safety or employment constraints, infer individual attendance or performance, operate systems or claim causality without a defensible design. Authorised owners approve and stop the pilot. If mapping, timing, coverage or sensor health is inadequate, exclude the stream; do not impute or substitute nearby data. The service cannot create, approve, publish or change the timetable.

WHY THIS IS USEFUL

Turns scenario planning into a measurable learning exercise rather than an unsupported rollout.

4. Catalogue summary and implementation requirements

The 80 questions define the evidence, authority, privacy and safe-failure capabilities required for responsible timetable decision support.

QUESTIONS

80

TT-001 to TT-080 across eight timetabling-centred categories.

R1 - PRACTICAL

11

Authoritative lookup, verified static information or current owner-issued status.

R2 - INTEGRATION-DEPENDENT

48

Cross-system integration and proposed privacy-preserving aggregate sensing.

R3 - RESEARCH OR RESTRICTED

21

Validated forecasting, optimisation, fairness analysis or restricted governance.

RECURRING CAPABILITY REQUIREMENTS

CONSTRAINT AND SOURCE RESOLUTION

Represent calendar, module, cohort, staff, room, access and policy constraints with owner, version, precedence, effective time, permission and conflict status.

ROUTE AND TRANSITION MODELLING

Model complete eligible routes, realistic travel and changeover intervals, lift or door dependencies and time-dependent restrictions; preserve route uncertainty.

DISRUPTION AND CHANGE PROPAGATION

Consume owner-issued closures and incidents, compare authorised alternatives and detect cross-system inconsistency without operating services.

ROOM, EQUIPMENT AND ACCESS SUITABILITY

Use current capacity by layout, equipment, whole-route accessibility, booking and service evidence. Vacancy or historical room use never creates eligibility.

PRIVACY-PRESERVING AGGREGATE FEEDBACK

Use anonymous room and route counts only for proportionate timetable review, with no identities, attendance histories, content or performance inference.

FAIRNESS, SCENARIOS AND ABSTENTION

Use explicit criteria and hard constraints, show trade-offs and sensitivity, protect restricted data and return not assessable when evidence is inadequate.

QUESTION RECORD CONTRACT

Every question maps a timetable decision and decision window to minimum authoritative constraints, justified proposed telemetry, verified spatial and temporal coverage, quality checks, a named operation, uncertainty, prohibited inference and safe failure.

5. Evidence, permissions and answerability rules

These catalogue-wide rules apply to every answer, regardless of readiness class or question category.

1. CURRENT CALENDAR AND CONSTRAINTS	Use current owner-issued calendar, module, cohort, staff, room and policy sources. Prior timetables and the May 2021 plan are context only.
2. CURRENT ROOM, ASSET AND SERVICE STATUS	Capacity, layout, equipment, access, booking, closures, maintenance and service health come from current owners. Telemetry never creates permission or eligibility.
3. PRIVACY AND MINIMUM NECESSARY DATA	Use aggregate cohort and occupancy evidence only where authorised. Do not expose identities, personal schedules, adjustment reasons, attendance, performance or productivity.
4. SPATIAL COVERAGE AND DATA QUALITY	Preserve room, route and asset mapping, sampling/aggregation, commissioning, calibration, clock alignment, maintenance, uptime, gaps, drift, relocation, provenance and source conflicts.
5. NO INDIVIDUAL ATTENDANCE OR PERFORMANCE INFERENCE	Anonymous counts may evaluate aggregate room use only. Do not reconstruct who attended, why, engagement, learning, staff performance or workload.
6. HARD CONSTRAINTS, FAIRNESS AND EXCEPTIONS	Academic, employment, accessibility, capacity, safety and room-owner constraints precede preferences. Fairness analysis uses explicit criteria and never ranks people.
7. ENVIRONMENTAL AND OCCUPANCY INTERPRETATION	Room-level numerical context may support qualified comparisons. It cannot certify comfort, ventilation, health, safety, availability or suitability by itself.
8. DISRUPTION, CLOSURES AND EMERGENCIES	Official closure, maintenance, security, certified alarm and emergency communications prevail. The service does not reopen rooms, suppress alarms or create emergency instructions.
9. OPERATION LABELS, PROVENANCE AND UNCERTAINTY	Separate authorised lookup, observation, calculation, estimate, forecast, comparison, diagnosis and recommendation. State source owner, times, controls, coverage and uncertainty.
10. TIMETABLE AUTHORITY AND PUBLICATION	The service supports draft decisions but does not create, approve, publish or change the authoritative timetable; grant rooms or access; operate systems; or override competent owners.
ANSWERABILITY RULE	When official evidence, permission or authority is missing, stale, conflicting, unauthorised or spatially insufficient, ask only the minimum necessary clarification, provide a qualified alternative where possible, defer to the responsible source, or return not assessable. Never conceal an evidence gap with an assumption or confident prose.